

INDUSTRIAL ELECTRONICS AND LABORATORY REPORT

The simulations completed in PSIM and the laboratory experiment will be presented.

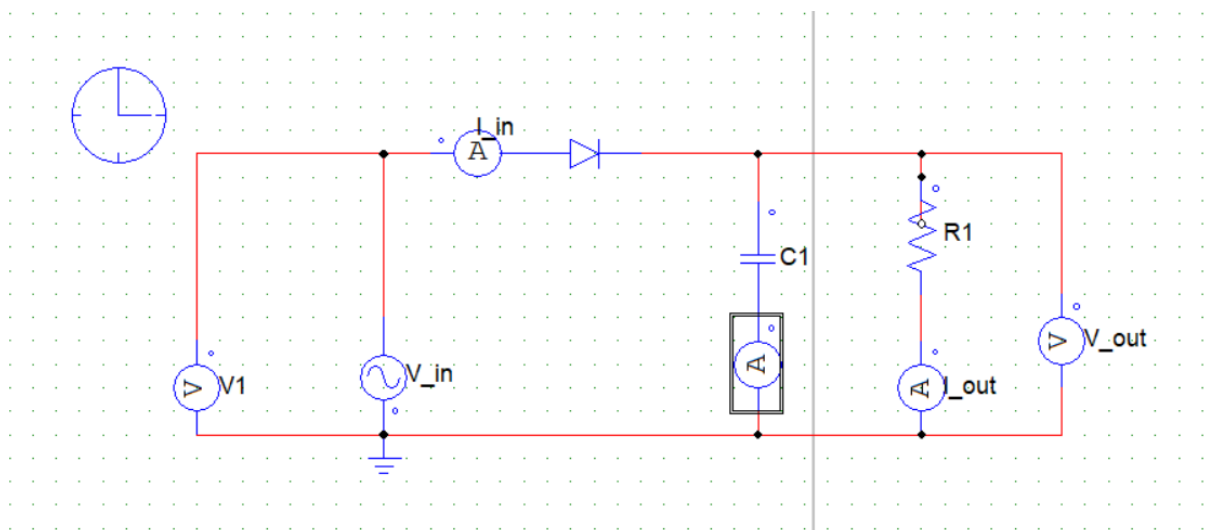
Name: Mustafa Yağcı

Matricola: 0830177

Rectifiers convert:

Rectifiers convert an input sinusoidal AC voltage into a unidirectional output voltage with a non-zero average value. However, this output has a pulsating nature and contains ripple, making it unsuitable for directly supplying electronic circuits. Therefore, filtering is required at both the input and the output. Output filters reduce voltage ripple and harmonics, while input filters improve the quality of the current drawn from the grid, ensuring that THD and power factor values meet standardized limits.

Single-Phase Half-Wave Rectifier:



The circuit consists of an input voltage $V(t) = V_s \sin(\omega t)$, a diode that blocks negative currents, and an RC output filter.

Given parameters are:

$$V_{\text{rms}} = 240 \text{ V, therefore } V_{\text{peak}} = V_{\text{rms}} \cdot \sqrt{2} = 340 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$R = 50 \, \Omega$$

$$\Delta Q = C \cdot \Delta V \rightarrow \Delta V = \frac{I_{avg} \cdot T_0}{C}$$

$$f_0 = 50 \text{ Hz} \rightarrow T_0 = 0.02 \text{ s}$$

$$T_0 \approx T_2$$

$$\Delta V = \frac{I_{avg}}{C} \cdot T_2 = V_0$$

The average current is the current flowing through the load and is proportional to the average voltage across it.

$$I_{avg} = \frac{V_{dc}}{R}$$

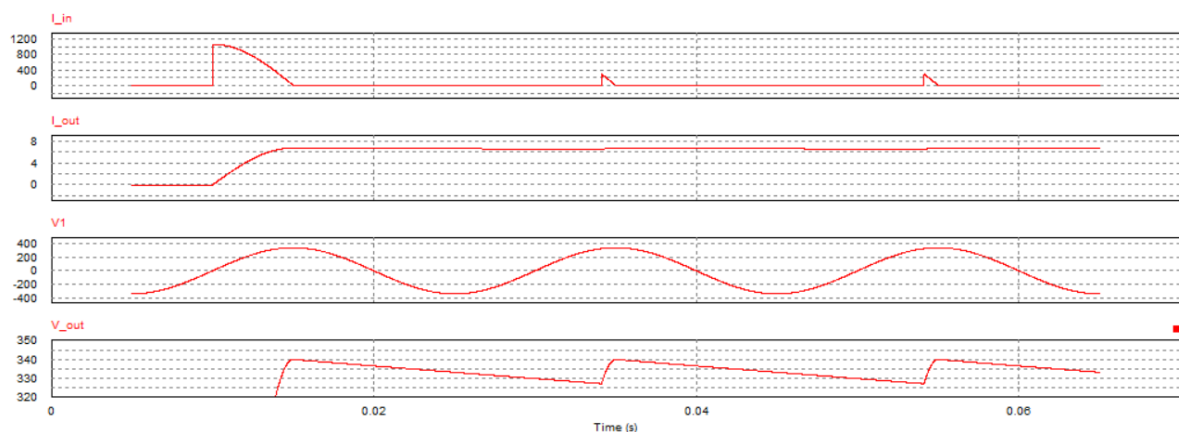
$$\Delta V = \frac{V_{dc}}{R \cdot C \cdot f_0}$$

$$V_{dc} \approx V_{0,max} \approx V_{0,avg}$$

$$\frac{\Delta V}{V_0} = \frac{1}{R \cdot C \cdot f_0} = 0.04$$

using this relation, we size the capacitor to obtain the desired ripple.

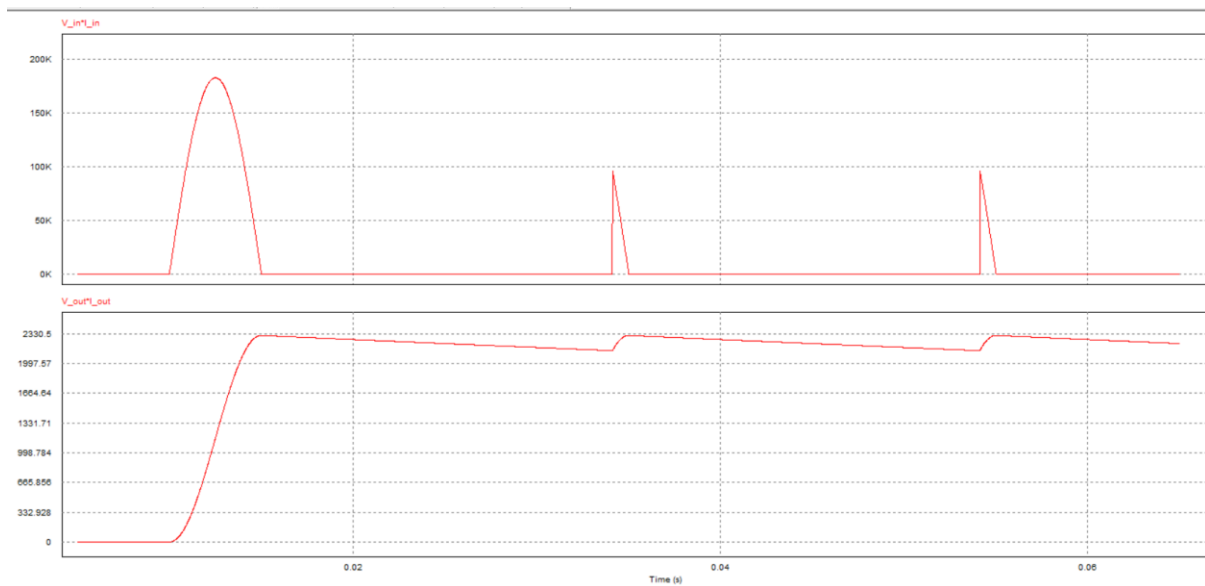
$$C = \frac{1}{f_0 \cdot R \cdot 0.04} = \frac{1}{50 \cdot 0.04} = 0.01 \text{ F} = 10 \text{ mF}$$



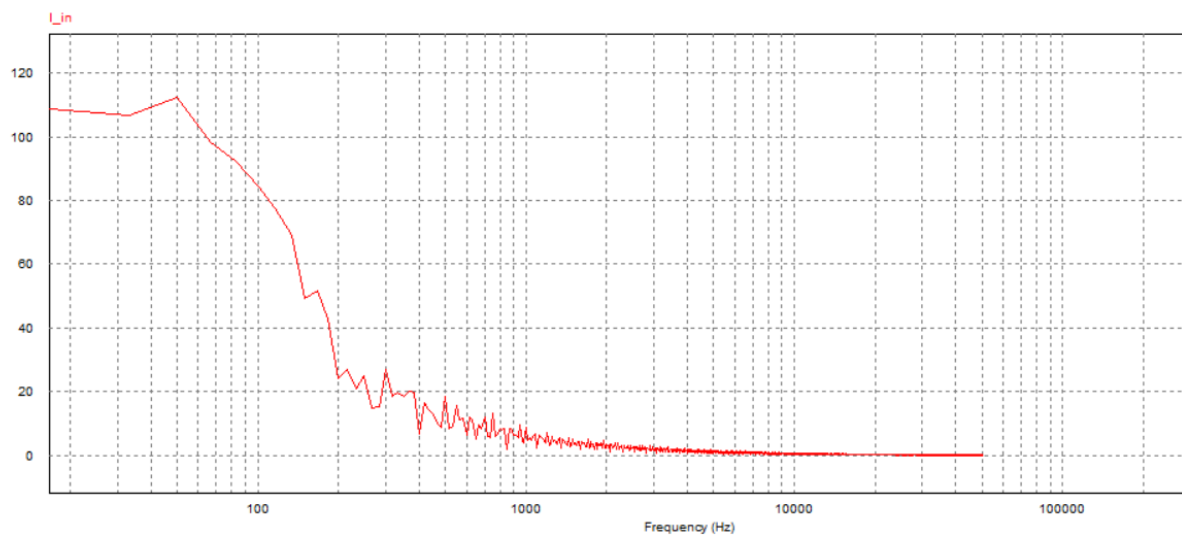
The output voltage follows the input voltage during the rising interval. In this period, the capacitor is charging, which is also indicated by the presence of positive current peaks. While the capacitor is charging, the diode conducts. When the input voltage drops below the capacitor voltage, the diode becomes reverse-biased and stops conducting.

This behavior is due to the large time constant, which prevents the capacitor from discharging quickly enough to fall below the input voltage. If the capacitor were not present, the output would simply be a positive half-wave identical to the input voltage.

During the capacitor discharge interval, both voltage and current can be approximated as varying linearly. The output current is proportional to the load conductance.



The input current exhibits a pulsating behavior, with peaks occurring during the conduction of the diode and the charging of the capacitor. A particularly high peak appears at the initial moment, since the capacitor is initially uncharged and must quickly follow the input voltage. As current is proportional to the derivative of the voltage, this results in a large current spike.



The graph shows the frequency spectrum of the input current. It can be observed that the harmonic at the fundamental frequency of 50 Hz has a greater amplitude compared to the other harmonics.

Handwritten calculations on grid paper:

ripple factor
 $V_{o_max} = 340\text{ V}$
 $V_{o_min} = 327\text{ V}$
 $\Delta V_o = 13\text{ V}$
 $\frac{\Delta V_o}{V_{o_median}} \approx 0.039$

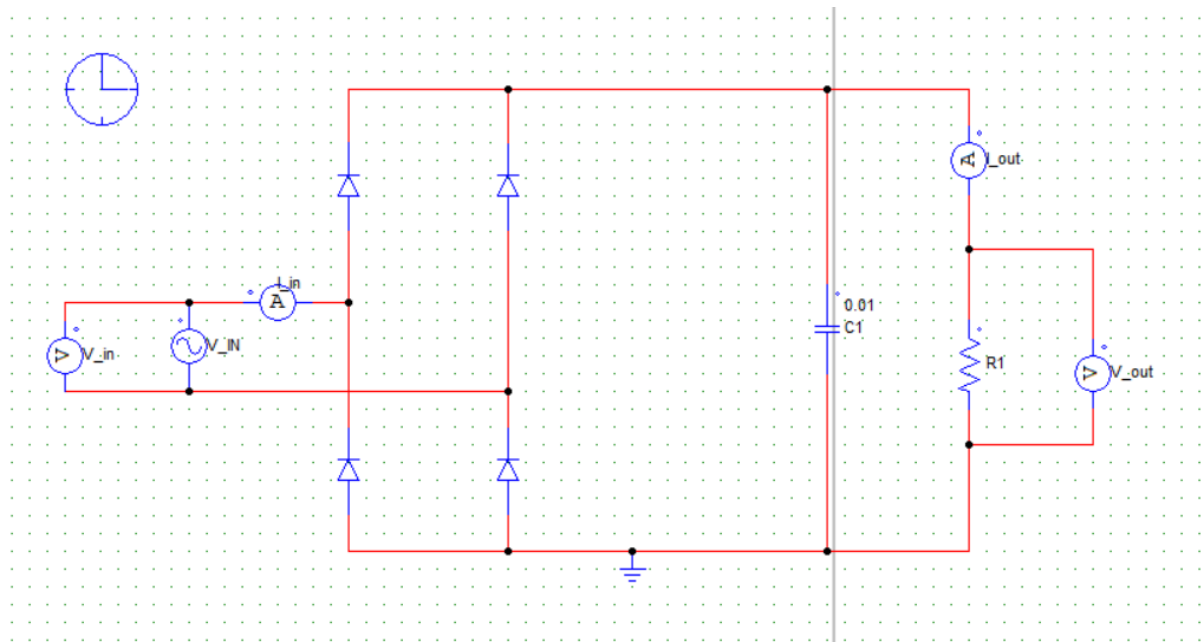
$\% \text{ THD} = \frac{I_{s1}}{I_s} \cdot 100 = 373\%$
 $\text{PF} = \frac{V_s \cdot I_{s1} \cdot \cos \phi}{V_s \cdot I_s}$
 $\text{PF} \approx 0.25$

The circuit has a THD of 373%, which is very high and makes it unsuitable for the power grid.

The displacement power factor (DPF) represents the phase difference between voltage and current. If DPF = 1, all power is effectively used. However, the power factor (PF) also considers current distortion. In this case, PF depends on both DPF and the distortion factor, which is related to THD.

A high THD reduces the distortion factor, leading to a low PF. Only the 50 Hz fundamental component contributes to real power, while higher harmonics cause losses in the circuit and the grid.

Single-Phase Full-Wave Rectifier:

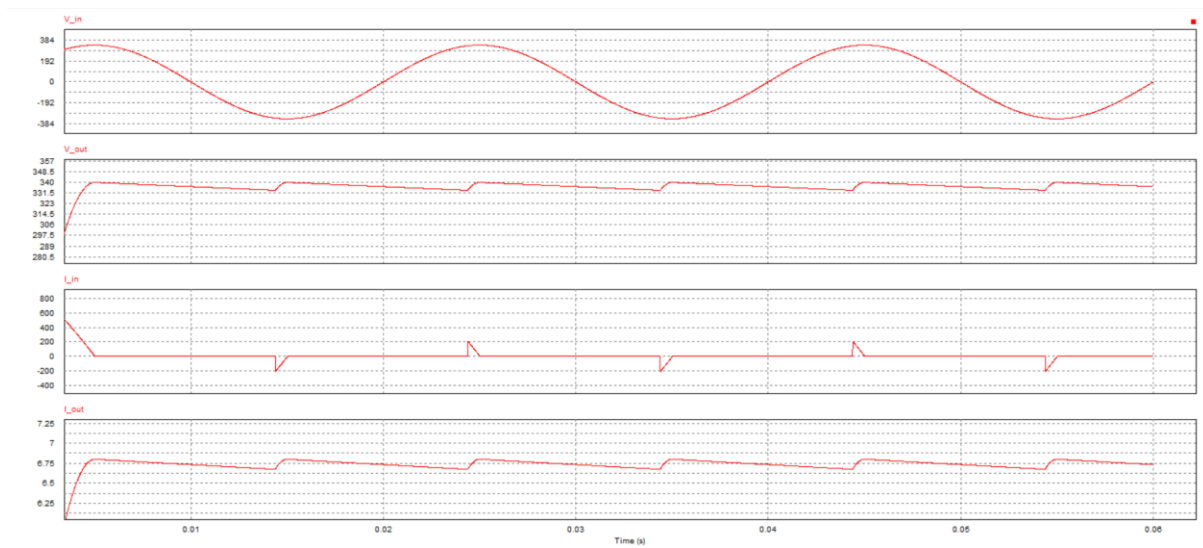


The full bridge rectifier consists of four diodes arranged in a bridge configuration. Unlike the half-wave rectifier, it produces a positive output voltage and current with a frequency twice that of the input.

This is achieved by providing an alternative path for the negative half-cycle, allowing current to flow and keeping the output always positive.

In the ideal case, the capacitor voltage follows the input during charging and behaves similarly to the half-wave case during discharging. Considering diode voltage drops, the total drop is higher since two diodes conduct simultaneously, requiring higher input voltage for good efficiency.

A key advantage of this rectifier is that it flips the negative half-cycle, resulting in two peaks per period. This doubles the frequency and allows the use of a smaller capacitor for the same ripple requirement, reducing design cost.

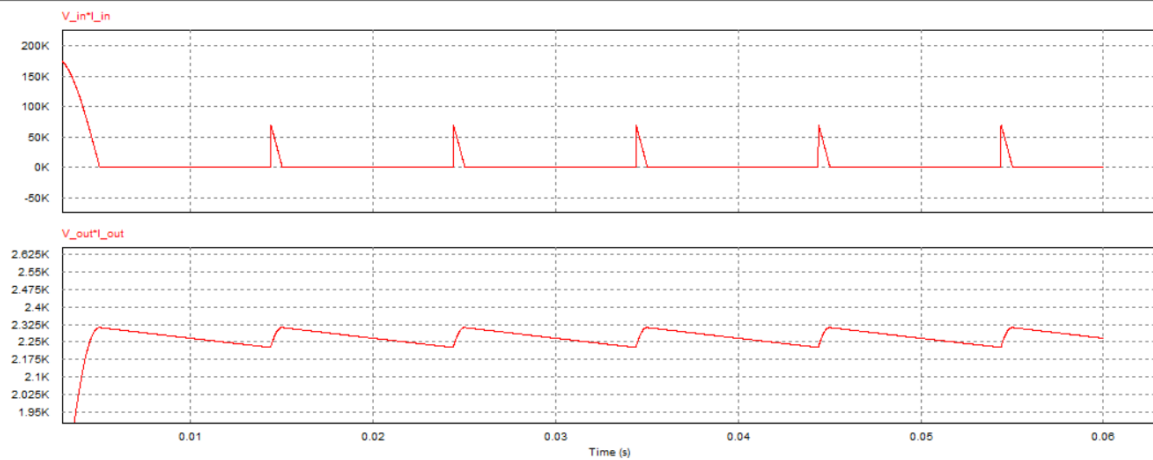


The initial peak is due to the capacitor charging during the transient phase. In steady state, smaller peaks with opposite polarity appear; these are also caused by the capacitor charging, but their area is smaller since less charge is transferred compared to the initial peak.

It can be observed that their frequency is doubled compared to the input voltage, as each negative half-cycle of the input is flipped by the diode bridge.

The input current is the sum of the currents flowing through the two diode branches. During positive voltage peaks, diodes 1 and 4 conduct, carrying positive current. During negative peaks, diodes 2 and 3 become forward-biased and conduct the negative current.

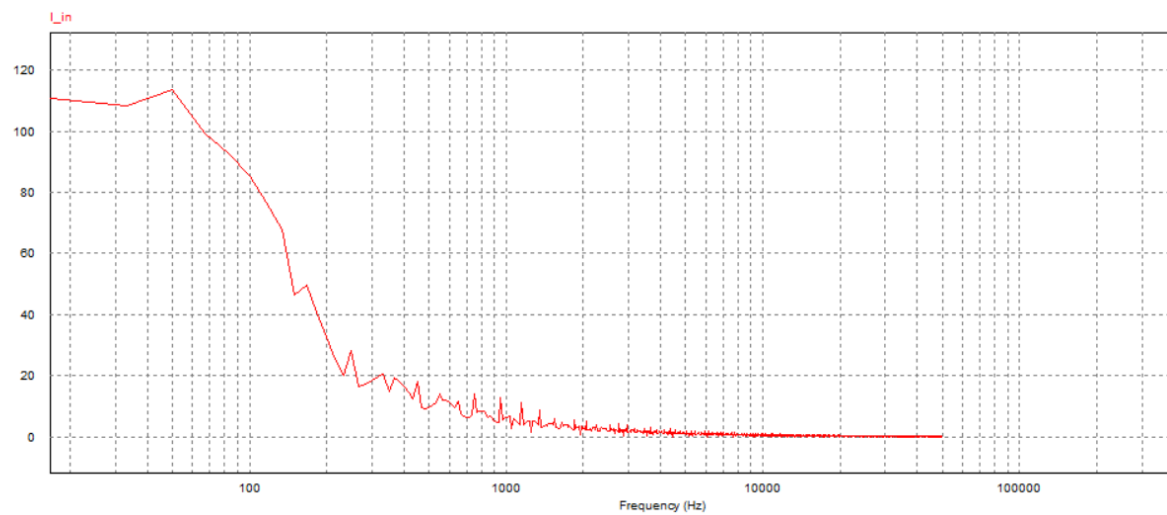
Below are the plots of the instantaneous input and output power. In the ideal case, with no voltage drops across the diodes and no parasitic resistances, the two powers are equal.

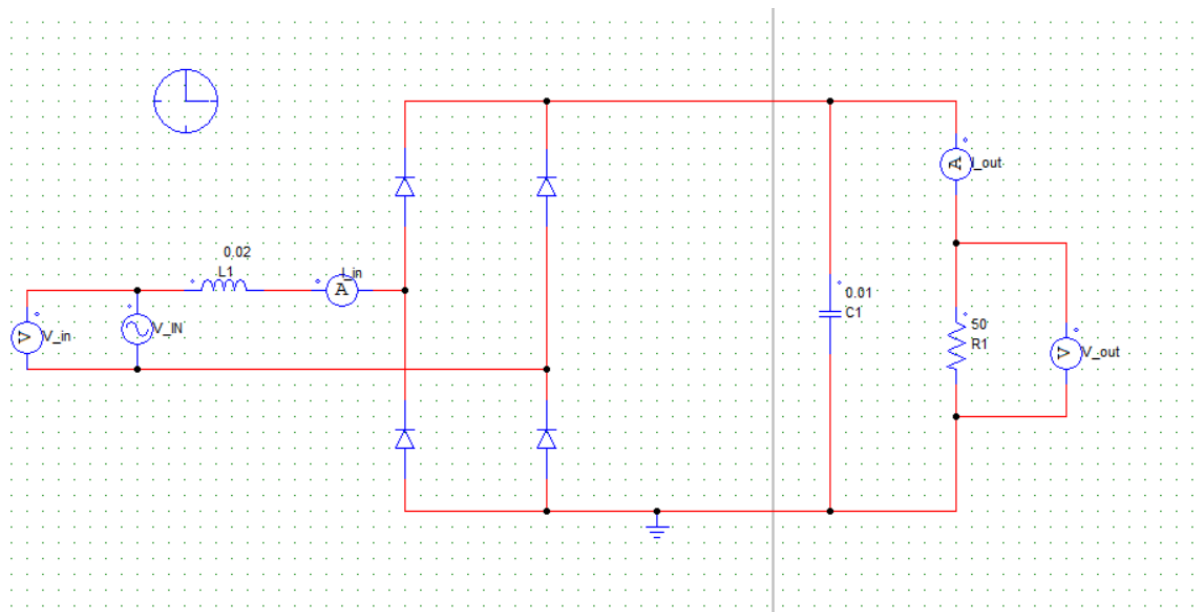


The following graph shows the FFT of the input current. Compared to the first circuit, the harmonic content is improved, with fewer peaks at frequencies above 100 Hz. This results in a lower THD, which is:

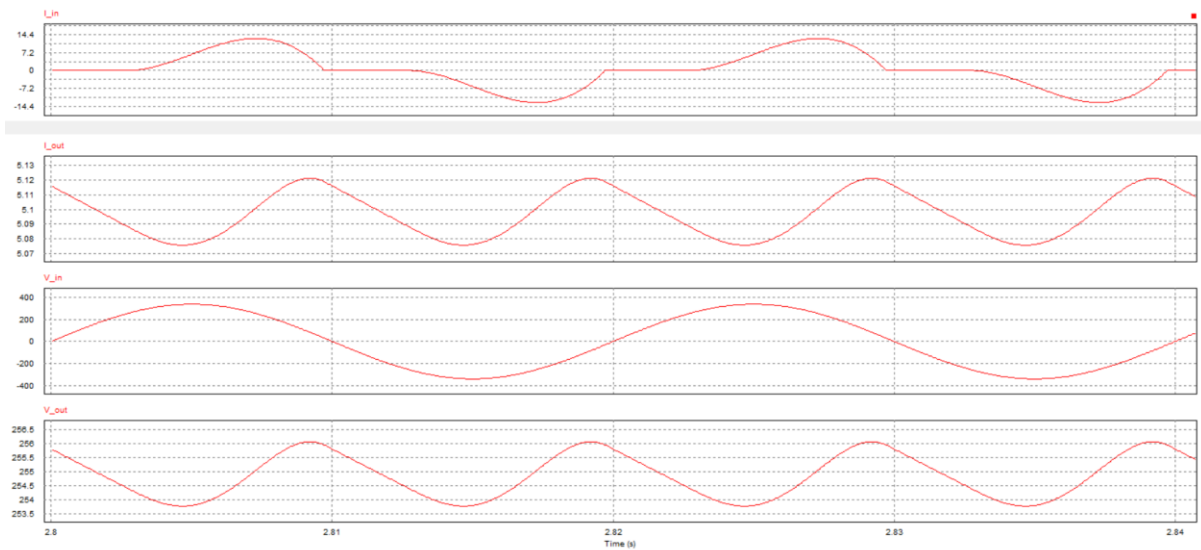
$$\text{THD}\% = 303\%$$

The power factor is $\text{PF} = 0.31$.





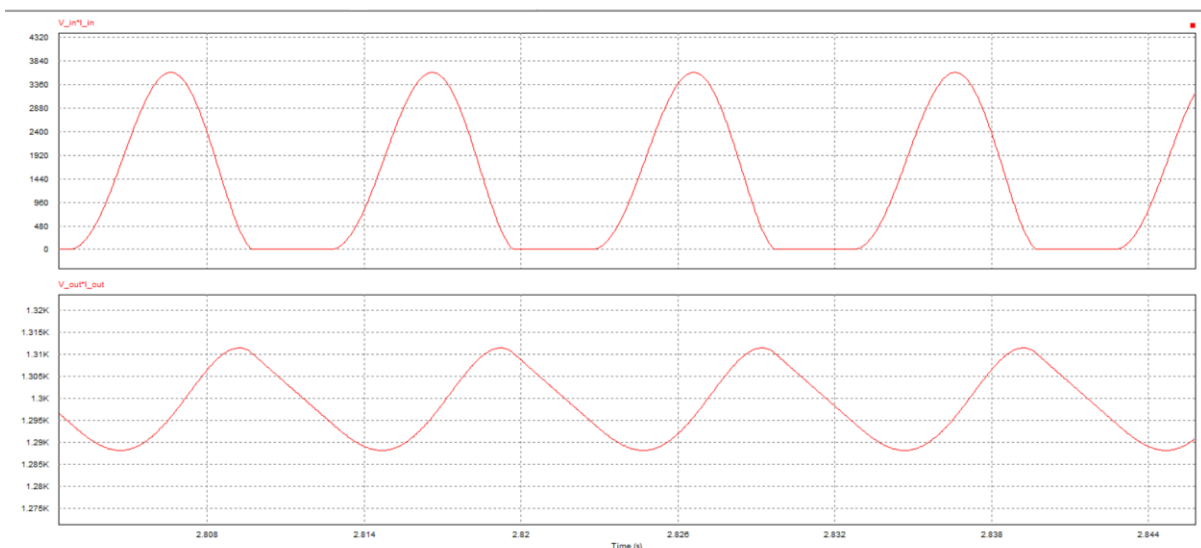
The same circuit is analyzed by adding an inductor $L = 20$ mH at the input. The presence of the inductor improves the current waveform by reducing distortion, since the current cannot vary abruptly.



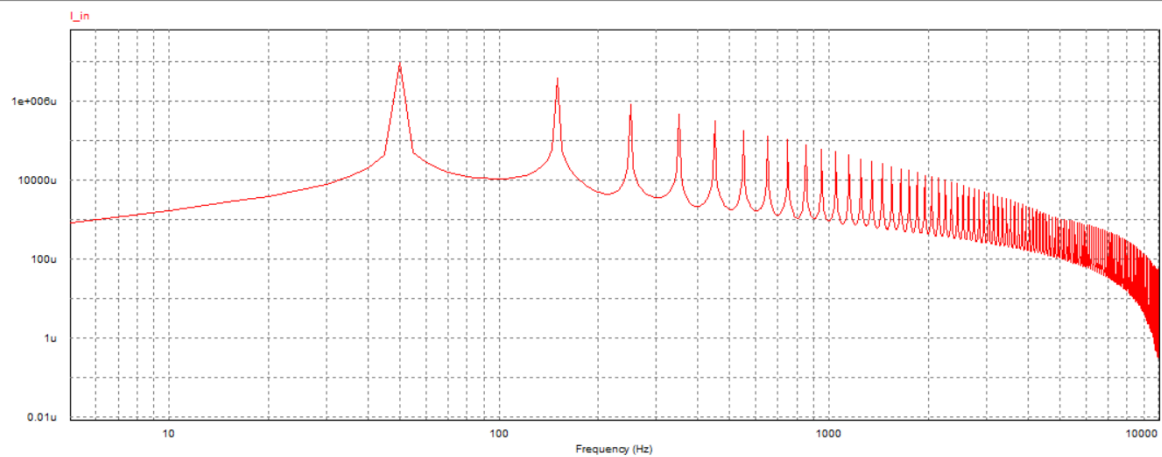
It can be observed that, for some intervals, the current has an almost sinusoidal shape.

However, it remains distorted and is zero for certain periods of time. This variation in current also affects the ripple.

Since the current cannot change instantaneously, the diodes conduct for a longer time, and the capacitor discharges for a shorter interval. This leads to a reduction in the difference between the maximum and minimum voltage across the capacitor, and therefore a reduction in ripple. Another consequence is a decrease in the DC component at the output and a slower transient response.



Also in this case, due to the absence of parasitic elements, the input power is equal to the output power. It can be observed that the input power remains positive in the intervals where current and voltage have the same sign.

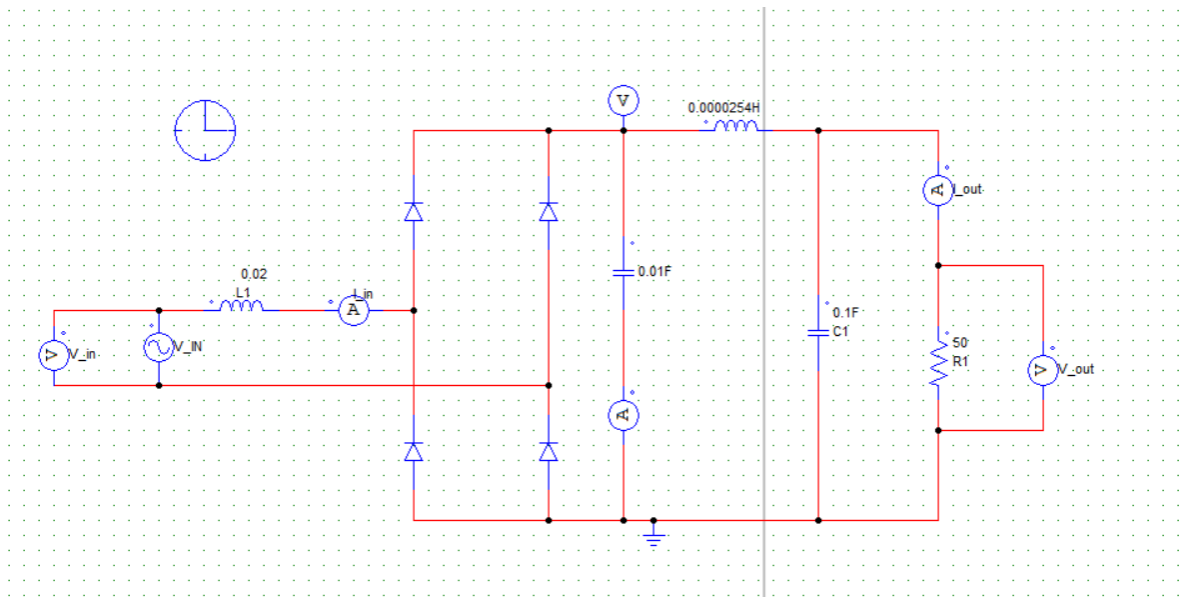


As mentioned before, the frequency spectrum improves due to the presence of the inductor. It can be considered cleaner in terms of harmonics compared to the previous cases, which results in a lower THD and a higher power factor.

THD% = 43.5%

PF=0.75

Single-Phase Full-Wave Rectifier with C-L-C Output Filter



In this case, a C-L-C filter is added at the output. The inductor and capacitors improve the filtering effect, reducing the ripple and making the output voltage smoother.

However, this filter does not affect the input side of the circuit. The diode bridge still sees a capacitor at its output, since the LC filter is placed after the rectifier and does not influence the harmonic content of the input current.

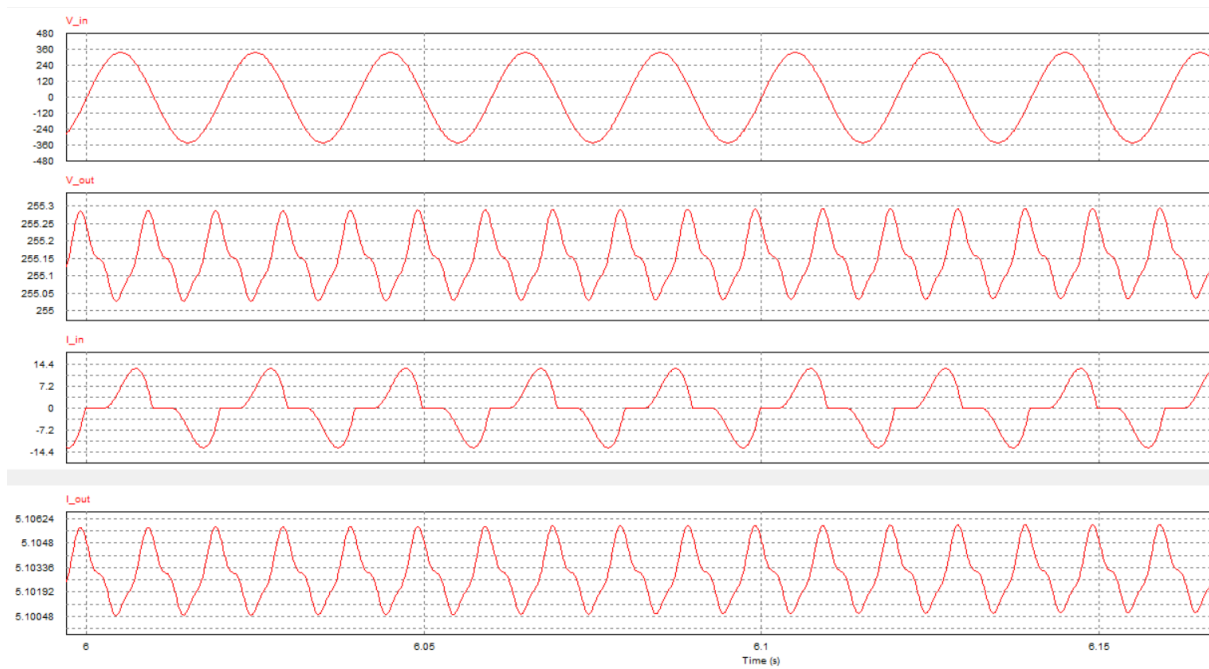
For this reason, THD and PF remain the same, while only the output voltage quality is improved.

$$L = \frac{1}{C_2 \cdot (f_c \cdot 2\pi)^2} = \frac{1}{0.1 \cdot (100 \cdot 2\pi)^2} = 2.53 \times 10^{-5} \text{ H}$$

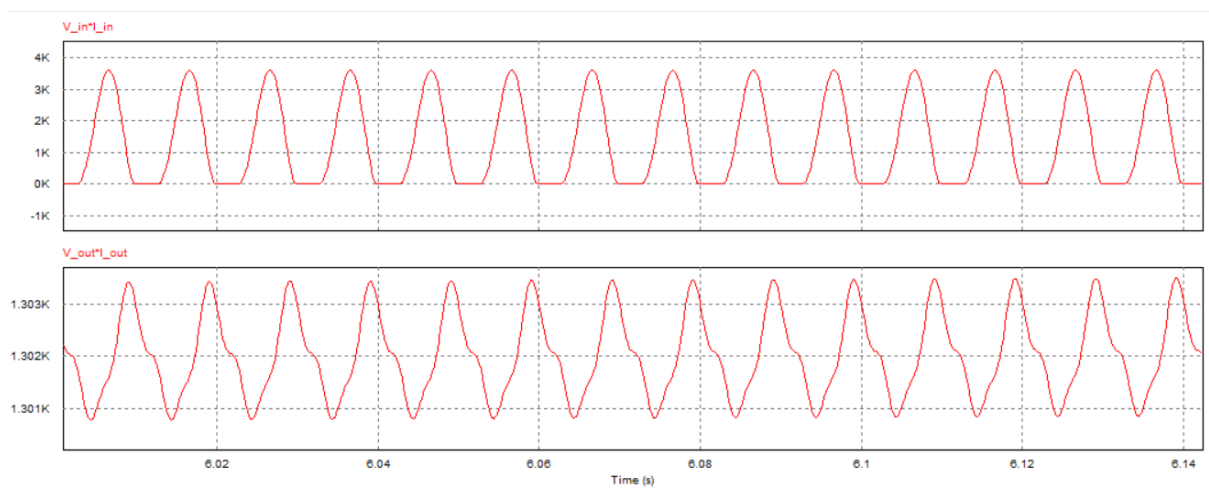
$$C_1 \leq C_2$$

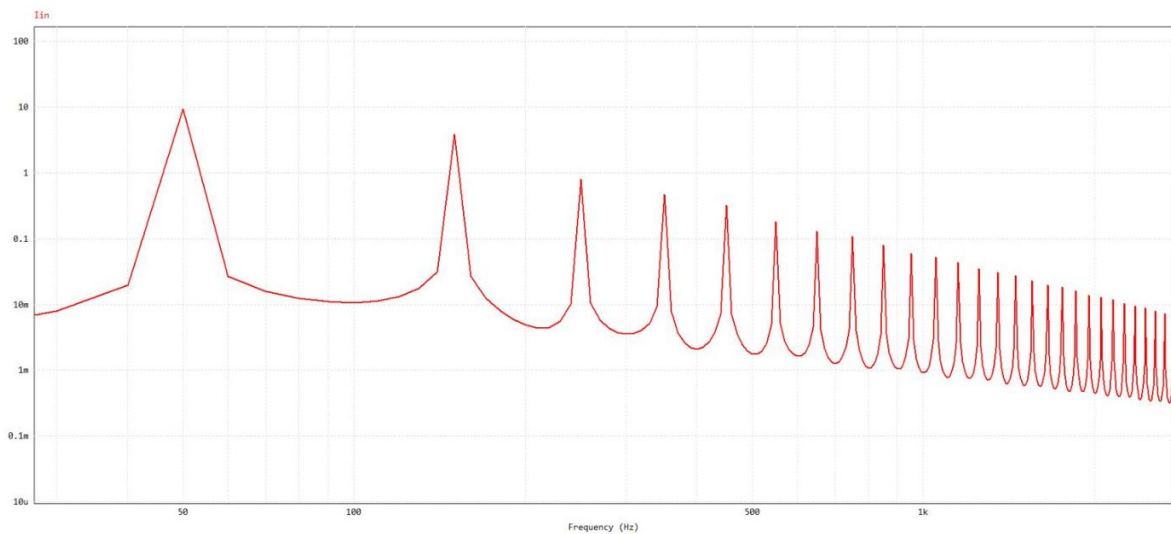
$$C_1 = \frac{C_2}{10}$$

$$C_1 = 10 \text{ mF} \quad C_2 = 0.1 \text{ F}$$



The input and output current and voltage values are provided.

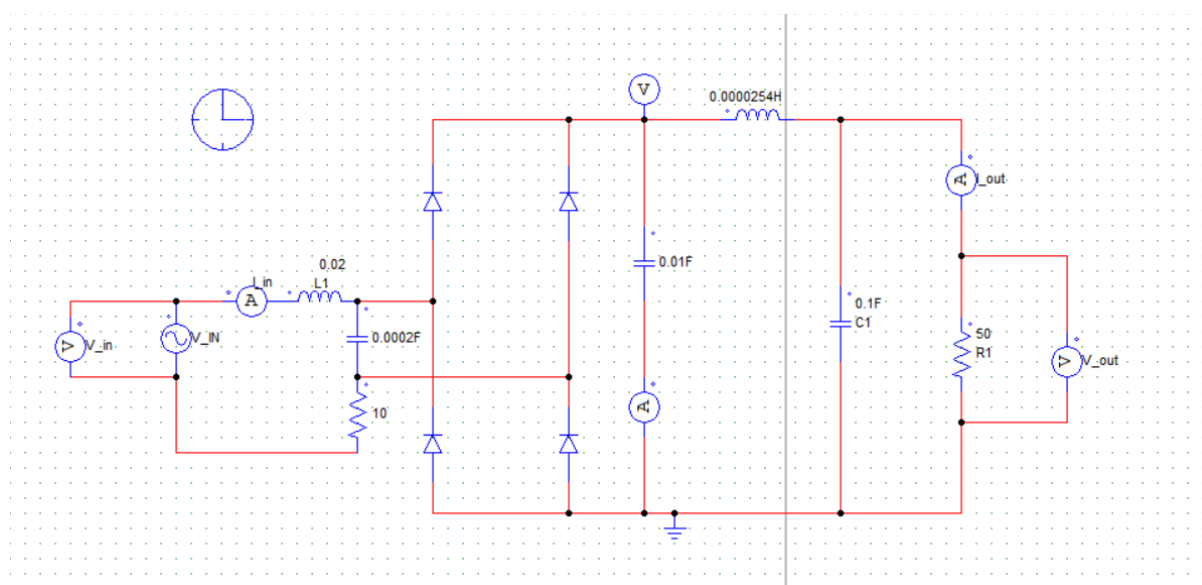




From the perspective of the input frequency spectrum, there are no changes compared to the previous case. This is because the input still sees a capacitor at the output of the diode bridge, while the LC filter is placed after the first capacitor.

The only improvement is in the output ripple. For this reason, THD and PF remain unchanged.

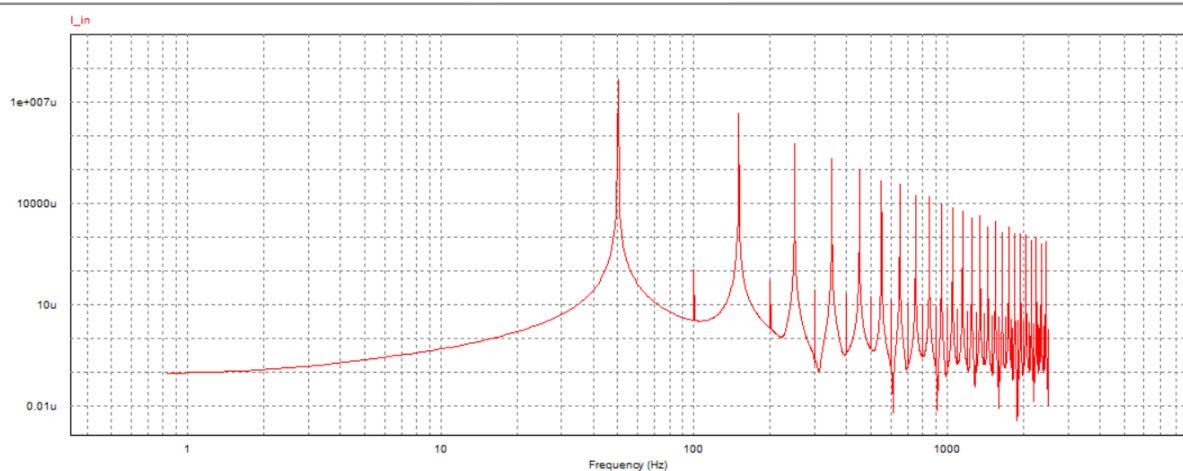
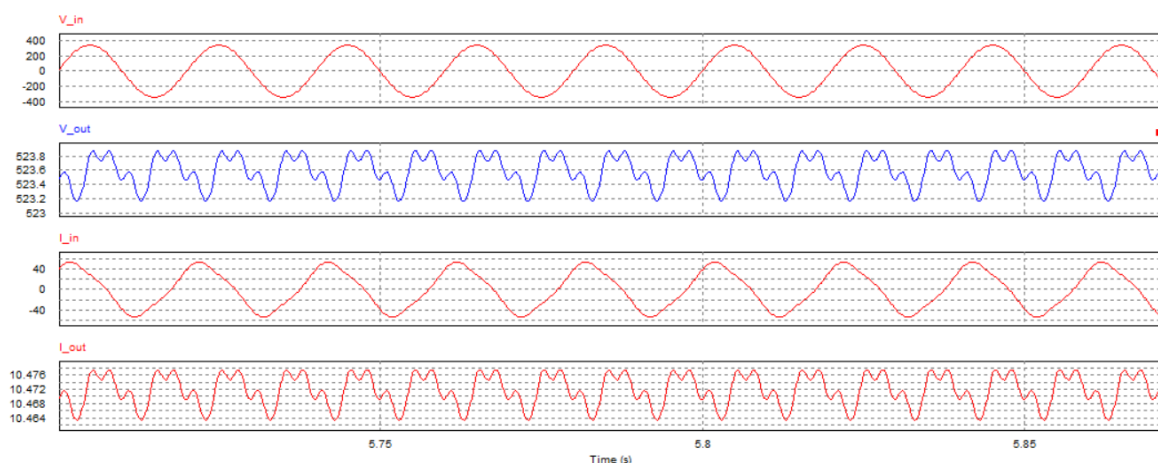
Single-Phase Full-Wave Rectifier with C-L-C Output Filter and L-C Input Filter:



The use of a properly designed input filter improves both the PF and THD of the circuit. Ideally, the goal is to achieve a power factor close to unity, which depends on both DPF and THD. By introducing an L-C filter designed with a cutoff frequency between the fundamental and the third harmonic, the THD is improved.

In the simulation, a resistor is added in series with the capacitor for two reasons. First, it modifies the quality factor Q of the filter. With zero resistance, the quality factor would be infinite, meaning that harmonics near the filter's natural frequency would be infinitely amplified. The damping resistor prevents this effect.

Second, it affects the phase of the impedance seen by the source. Without the resistor, current and voltage are phase-shifted, with the current leading the voltage by 90 degrees in the capacitor. By adding the resistor, the phase of the equivalent impedance is reduced, improving the power factor.



I want the cutoff frequency (ω_c) to be lower than the third harmonic at 150 Hz so that components from the third harmonic and above can be filtered out.

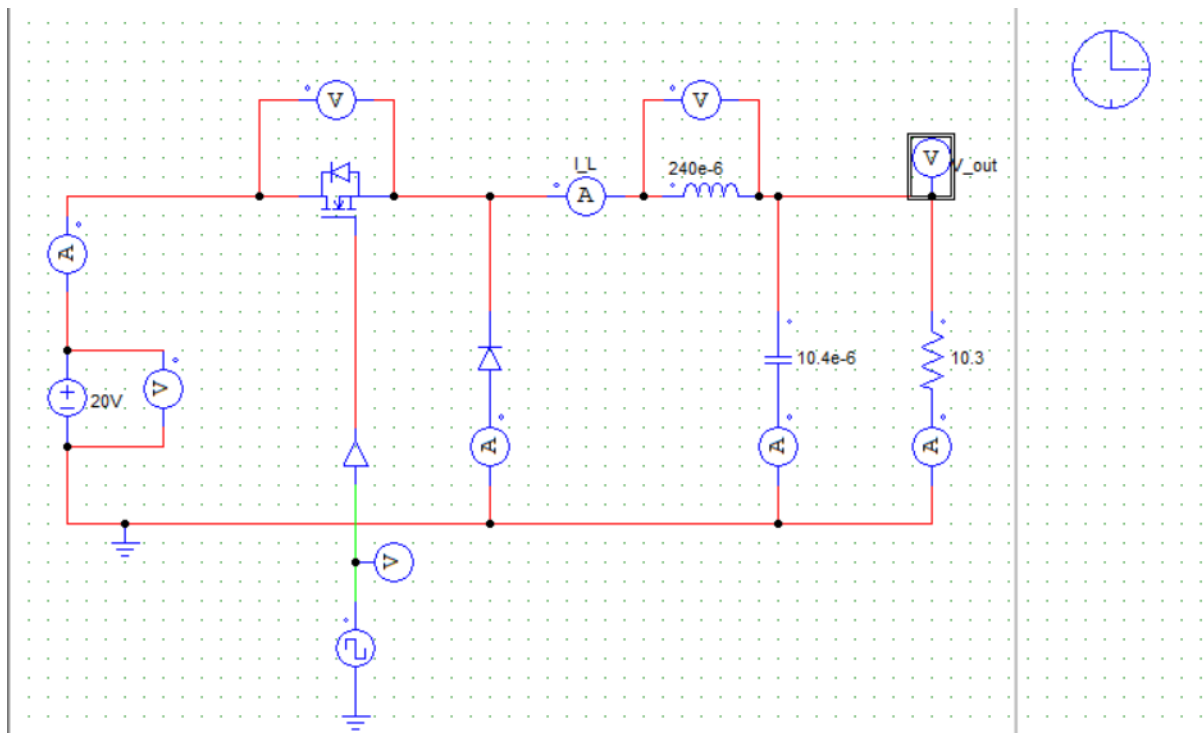
Additionally, I set $\omega_c = 80$ Hz so that it does not attenuate the 50 Hz input sinusoidal signal.

$$L = 0.02$$

$$C = \frac{1}{L \cdot (f_c \cdot 2\pi)^2} = \frac{1}{0.02 \cdot (80 \cdot 2\pi)^2} \approx 2 \cdot 10^{-4}$$

$$R = \sqrt{\frac{L}{C \cdot \pi}} = \sqrt{\frac{0.02}{2 \cdot 10^{-4}}} = 10$$

CONVERTITORE DC/DC BUCK:



This buck converter is designed such that the input DC voltage is switched using a PWM-controlled switch with a defined duty cycle. When the switch is on, energy is stored in the inductor; when it is off, the inductor current continues to flow through the freewheeling diode. The inductor and output capacitor work together to reduce current and voltage ripple, providing a smoother DC output. The output voltage is controlled by adjusting the duty cycle.

The following section includes the sizing of the components, the calculation of key parameters, and a comparison with the values obtained from the simulation.

$$\Delta I_L = \frac{1}{L} (V_{in} - V_o) \cdot D \cdot T_s = \frac{8 \times 30}{240} = 1A$$

$$f_s = 20kHz \quad T_s = 50\mu s$$

$$T_{on} = D \cdot T_s = 20\mu s$$

$$V_o = D \cdot V_{in} = 12V$$

$$V_L = (V_{in} - V_o) = 8V$$

$$P_o/V_o = I_o = 1,167A$$

$$I_{in} = D \cdot 1,167 = 0,7A$$

$$I_{Lmax} = I_o + \frac{1}{2} \Delta I_L = 1,167 + 0,5 = 1,667A$$

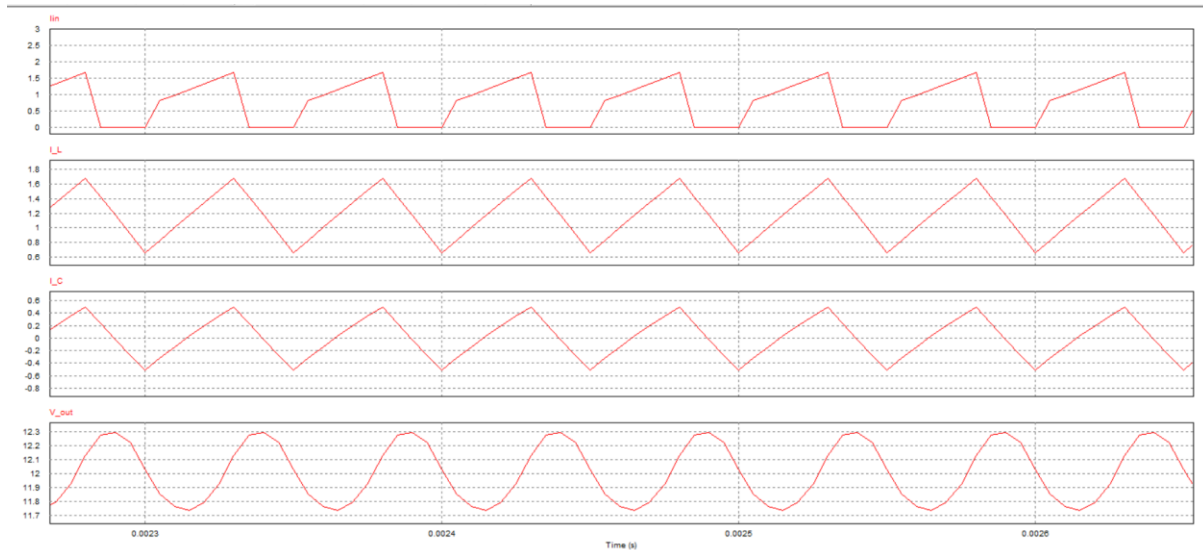
$$I_{Lmin} = I_o - \frac{1}{2} \Delta I_L = 1,167 - 0,5 = 0,667A$$

$$\frac{\Delta V_o}{V_o} = \frac{1}{8} \frac{(1-D)}{L \cdot C \cdot f_s^2} = 0,05$$

$$C = \frac{(1-D)}{8 \cdot 0,05 \cdot L \cdot f_s^2} = \frac{0,4}{8 \cdot 0,05 \cdot 240 \times 10^{-6} \cdot (20 \cdot 10^3)^2} = 10,4 \times 10^{-6} F$$

$$P_o = \frac{V_o^2}{R} \rightarrow R = \frac{V_o^2}{P_o} = \frac{12^2}{14} = 10,2\Omega$$

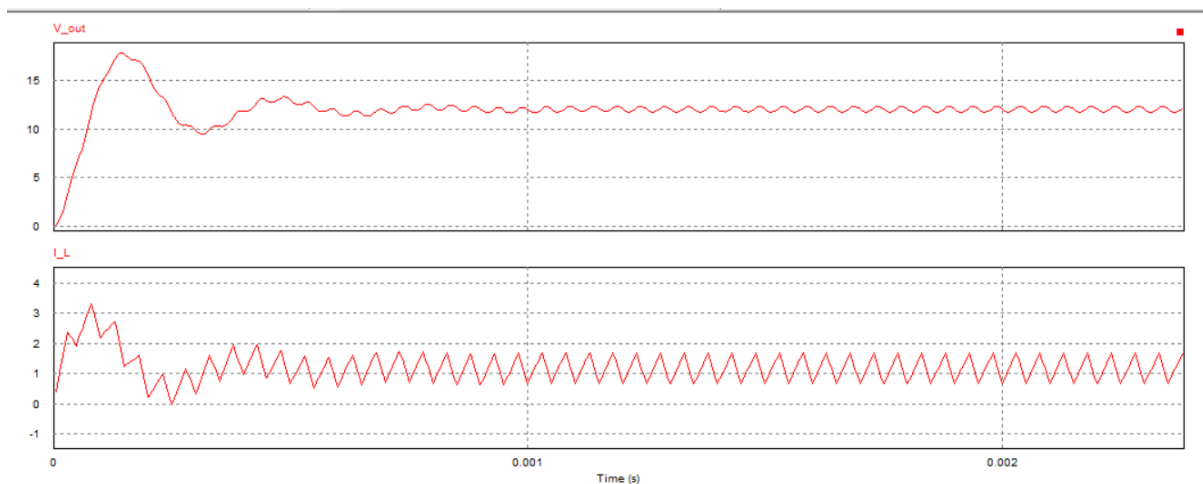
$$I_o = \frac{P_o}{V_o} = \frac{14}{12} = 1,17A$$



As expected, the input current is non-zero only when the switch conducts, and its average value is proportional to the duty cycle. The inductor current has a triangular shape: it increases when the switch is ON and decreases when it is OFF.

This current consists of a DC component (average load current) and a ripple component that depends on the switching frequency and inductance. The output voltage has a low ripple, composed of a DC value and a high-frequency component due to switching noise.

The graphs represent steady-state behavior. The transient response depends on the second-order nature of the circuit and varies with the load resistance.



The system has an underdamped response, so voltage and current peaks appear due to the inductor and capacitor.

The power switch must handle both the voltage across it during the OFF state and the peak inductor current during the ON state. Similarly, the load and the capacitor must withstand the peak output voltage.

Since the inductance value is **103 μH** ,

a value of **50 μH** was selected for DCM mode and the results were observed.

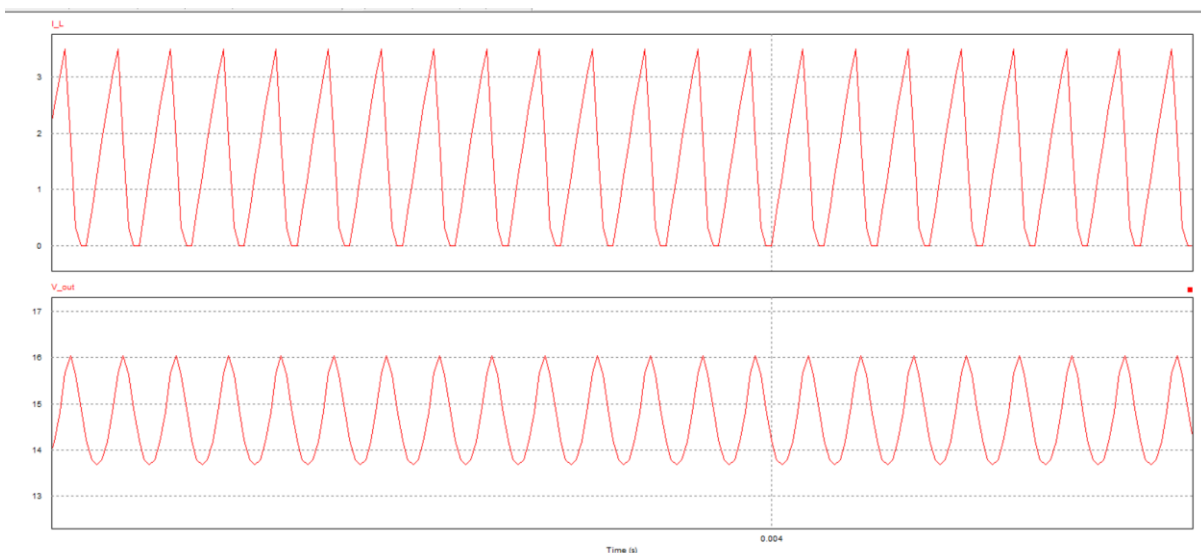
$$L_c = \frac{R \cdot (1-D)}{2 f_s} = \frac{103 \cdot (0.4)}{2 \cdot 20 \cdot 10^3} = 103 \mu\text{H}$$

$$L_2 = 50 \mu\text{H}$$

$$I_{L,p} = 3.44 \text{ A}$$

$$I_{L,m} = 3.54 \text{ A}$$

$$V_o = 1.48 \text{ V}$$



This makes the circuit respond faster. However, the inductor current becomes zero for some time intervals, meaning the converter operates in DCM. In this mode, the current ripple and peak values are higher compared to CCM.

DCM has some advantages such as lower cost, faster response, and reduced switching losses. However, it also has disadvantages: higher ripple, more complex behavior, and a nonlinear gain, which makes control more difficult.

Next, the same circuit is analyzed by including parasitic effects such as voltage drops and resistances. Components from different manufacturers were used.

Electronic components used:

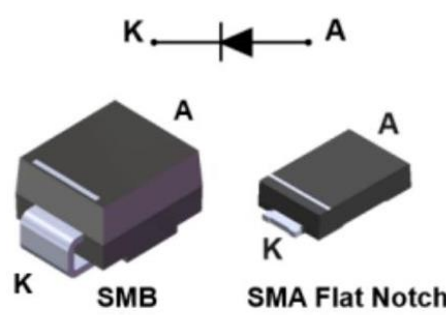
MOSFET: N-Channel 30 V

Values taken from the MOSFET datasheet: maximum current 6 A, maximum Drain-Source voltage 30 V (10 V in our case), $R_{DS(on)}$, and rise and fall delay times used to calculate switching losses.

Dynamic ^b						
Input capacitance	C_{iss}	$V_{DS} = 15\text{ V}, V_{GS} = 0\text{ V}, f = 1\text{ MHz}$	-	424	-	pF
Output capacitance	C_{oss}		-	100	-	
Reverse transfer capacitance	C_{rss}		-	42	-	
Total gate charge	Q_g	$V_{DS} = 15\text{ V}, V_{GS} = 10\text{ V}, I_D = 5.5\text{ A}$	-	8.2	13	nC
Gate-source charge	Q_{gs}	$V_{DS} = 15\text{ V}, V_{GS} = 4.5\text{ V}, I_D = 5.5\text{ A}$	-	4.2	7	
Gate-drain charge	Q_{gd}		-	1.4	-	
Gate resistance	R_g		-	1.4	-	
Gate resistance	R_g	$f = 1\text{ MHz}$	2.5	12.6	25.2	Ω
Turn-on delay time	$t_{d(on)}$	$V_{DD} = 15\text{ V}, R_L = 3.4\ \Omega$ $I_D \cong 4.4\text{ A}, V_{GEN} = 4.5\text{ V}, R_g = 1\ \Omega$	-	6	12	ns
Rise time	t_r		-	20	30	
Turn-off delay time	$t_{d(off)}$		-	14	21	
Fall time	t_f		-	10	20	
Turn-on delay time	$t_{d(on)}$	$V_{DD} = 15\text{ V}, R_L = 3.4\ \Omega$ $I_D \cong 4.4\text{ A}, V_{GEN} = 10\text{ V}, R_g = 1\ \Omega$	-	3	6	
Rise time	t_r		-	11	20	
Turn-off delay time	$t_{d(off)}$		-	20	30	
Fall time	t_f		-	7	14	

PRODUCT SUMMARY	
V_{DS} (V)	30
$R_{DS(on)}$ max. (Ω) at $V_{GS} = 10\text{ V}$	0.028
$R_{DS(on)}$ max. (Ω) at $V_{GS} = 4.5\text{ V}$	0.033
Q_g typ. (nC)	4.2
I_D (A) ^{a, e}	6
Configuration	Single

DIODE: STPS2L25. Schottky diode with a very low forward voltage drop.



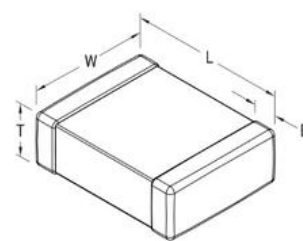
The diagram shows the diode symbol with cathode (K) and anode (A) terminals. Below it are two physical packages: an SMB package and an SMA Flat Notch package, both with K and A terminals labeled.

Product status	
STPS2L25	

Product summary	
Symbol	Value
$I_{F(AV)}$	2 A
V_{RRM}	25 V
$T_{J(max.)}$	150 °C
$V_F(typ.)$	0.325 V

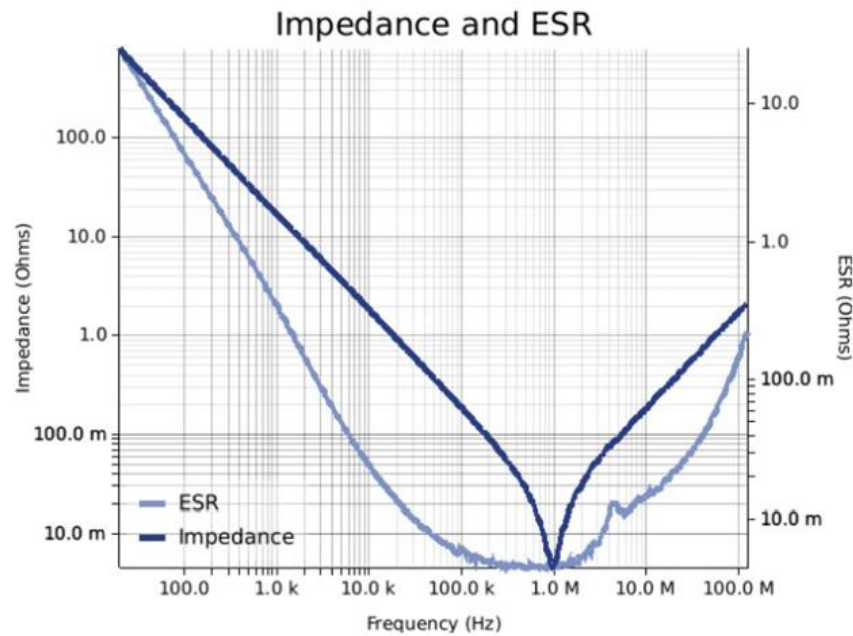
CAPACITOR: CC1206KFX7R8BB106.

Approximately, $R_{esr}=10\text{m Ohm}$

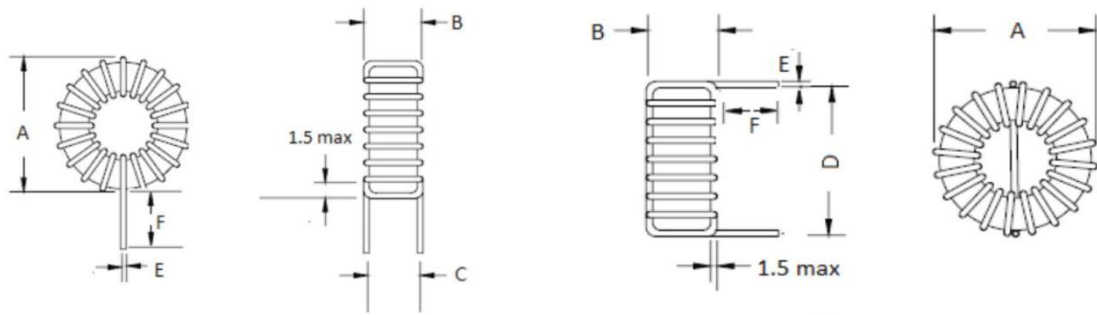


A 3D perspective drawing of a rectangular capacitor with dimensions L (length), W (width), and B (height) indicated.

Specifications	
Capacitance	10 μF
Tolerance	10%
Voltage DC	25 VDC
Temperature Range	-55/+125°C



INDUCTOR: ATCA-04-271M

L=270mH, R=0.081ohm, I_{DC}=3A, Classe di tolleranza M=±20%


	Vertical	Horizontal
ATCA-04-271M	270	M
	0.081	3.0
		0.80

Considering all relevant parasitic elements and the MOSFET's rise and fall delay times, the converter's losses and efficiency were calculated.

- Mosfet conduction Losses

$$P_{C,loss} = I_o^2 \cdot R_{DS(on)} \cdot D \cdot \frac{1}{f_s} = 1,17^2 \cdot 0,028 \cdot 0,6 = 22 \cdot 10^{-3} \text{ W}$$

- Switching loss =

$$P_{S,loss} = \frac{1}{2} \cdot V_{DS} \cdot I_o \cdot (t_{on} + t_{off}) \cdot f_s = 536 \cdot 10^{-3} \text{ W}$$

$t_{on} = 3 \text{ ns}$
 $t_{off} = 20 \text{ ns}$

$$P_{mosfet} = P_C + P_S = (5,6 + 22) \times 10^{-3} = 27,4 \text{ mW}$$

- Inductor losses

$$P_L = I_o^2 \cdot R_L = 1,17^2 \cdot 0,081 = 0,11 \text{ W}$$

- Capacitor loss

$$P_C = I_o^2 \cdot R_C \approx 0$$

- Diode loss

$$P_{D,loss} = I_o \cdot V_{th}(1-D) = 1,17 \cdot 0,325 \cdot 0,4 = 0,152 \text{ W}$$

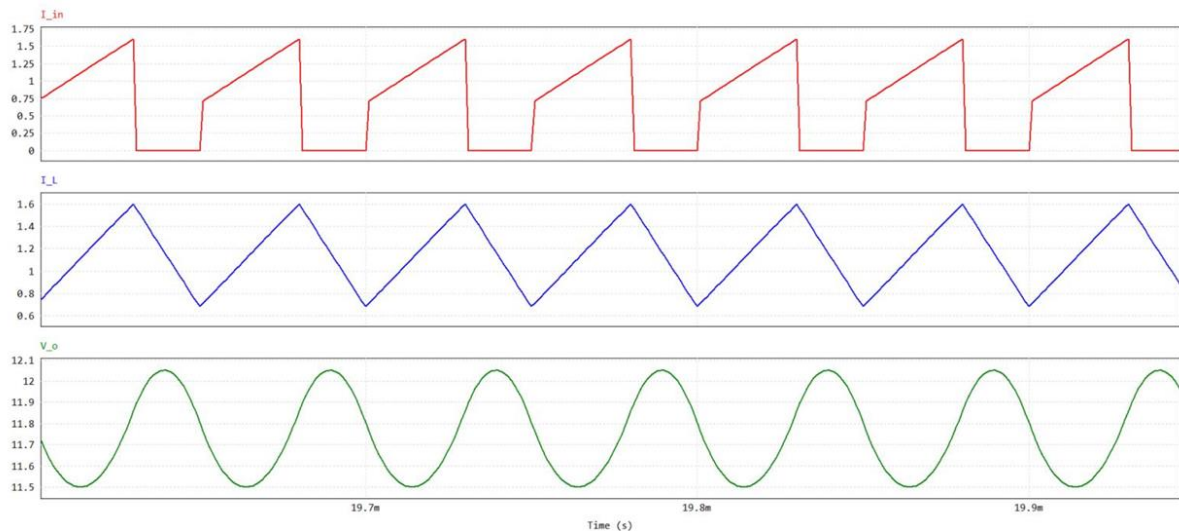
- Total power loss

$$P_{T,loss} = P_{mos} + P_{L,loss} + P_{C,loss} + P_{D,loss} \approx 290 \text{ mW}$$

- Efficiency =

$$\frac{P_o}{P_i} = \eta \quad P_i = P_o + P_{T,loss} \quad \eta = \frac{P_o}{P_o + P_{T,loss}} = \frac{14}{14,290} = 0,98 \quad \eta \% = 98 \%$$

The following graphs show the results obtained from the simulation after inserting the datasheet values.



The waveform shapes remain more or less the same, but the values are different:

Average input current: $I_{in,avg} = 0.694 \text{ A}$

Inductor current:

Average: $I_{L,avg} = 1.14 \text{ A}$

Peak: $I_{L,peak} = 1.6 \text{ A}$

Minimum: $I_{L,min} = 0.68 \text{ A}$

Ripple: $\Delta I_L = 0.92 \text{ A}$

Output voltage:

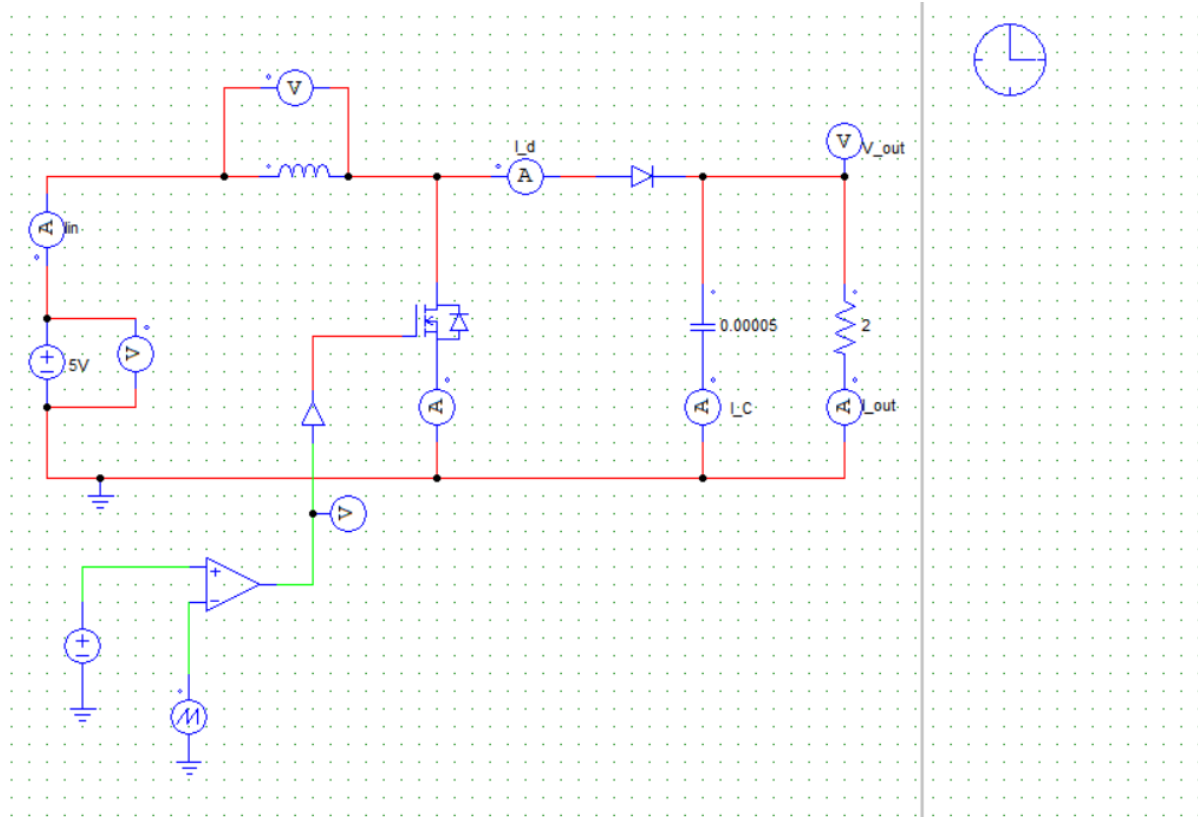
Average: $V_{o,avg} = 11.7 \text{ V}$

Ripple: $\Delta V_o = 0.05 \text{ V}$

$$\frac{V_o}{V_d} = D \cdot \frac{R + r_c}{R + r_c + r_L} = 0.5 \cdot \frac{10.3 + 0.01}{10.3 + 0.01 + 0.081} = 0.59$$

updated output voltage =

$$V_o \approx 11.8 \text{ V}$$

BOOS CONVERTER DC/DC:

This converter increases the output voltage, so the output voltage is higher than the input, and the input current is higher than the output current.

The inductor current has a triangular shape. If the converter works in CCM, the current never becomes zero, which means the input current is smoother and THD is improved.

The circuit includes an inductor (stores energy), a diode (provides current when the switch is off), a PWM-controlled switch (controls the output voltage), and a capacitor (reduces voltage ripple). The boost converter has a gain greater than 1. Since there is no inductor at the output, the voltage ripple is not affected by it.

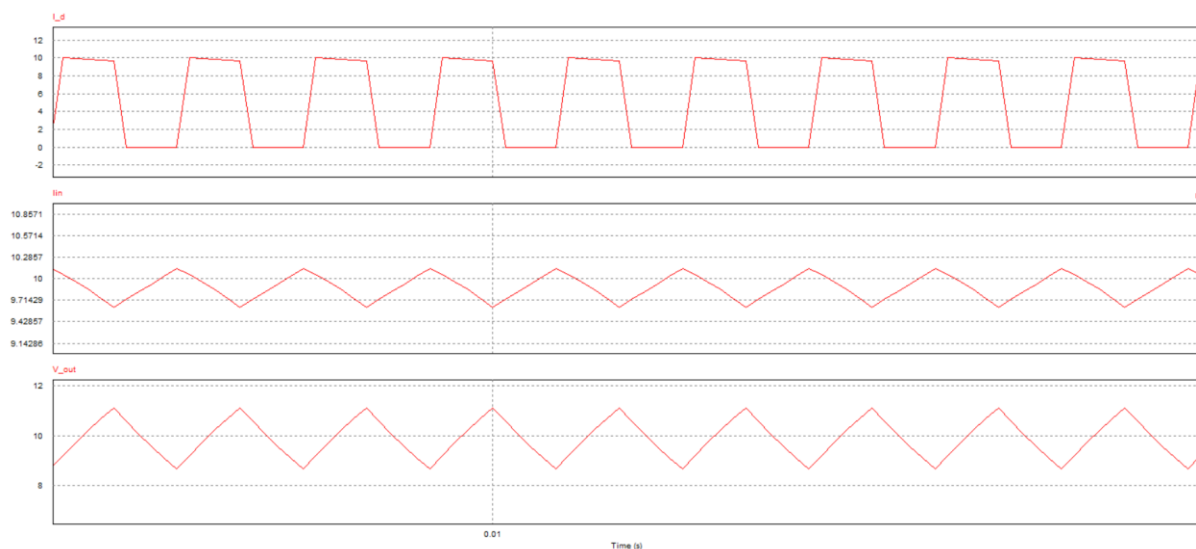
$$\begin{aligned}
 V_D &= 5V & V_{out} &= 10V & P_{out} &= 50W & F_s &= 20kHz \\
 L &= 240\mu H \\
 V_o &= V_D \frac{1}{1-D} \rightarrow D = \frac{V_D - V_o}{-V_o} & D &= \frac{5}{10} = 0.5 \\
 P_o &= \frac{V^2}{R} \rightarrow R = \frac{V^2}{P_o} = \frac{100}{50} = 2\Omega \\
 \frac{\Delta V_o}{V_o} &= \frac{D \cdot T_s}{R_c} = \frac{0.5 \cdot (20 \cdot 10^{-3})^{-1}}{2 \cdot C} & C &= \frac{0.5 \cdot 5 \cdot 10^{-5}}{2 \cdot 0.25} = 50\mu F
 \end{aligned}$$

The graphs show the inductor current (same as input current) and the diode current. The diode current has both a DC and a filtered AC component.

When the switch is OFF, the inductor charges the capacitor, so the capacitor voltage increases almost linearly. When the switch is ON, the capacitor supplies energy to the load, so its voltage decreases almost linearly.

Because of this, the output voltage has a triangular shape instead of a sinusoidal one.

The ripple depends on the capacitor value. It is calculated by looking at how much charge the capacitor gains during OFF time and loses during ON time.



inductor analysis =

During the switch ON state =

$$V_L = L \cdot \frac{di}{dt} \rightarrow \Delta I_L = \frac{V_L}{L} \cdot D \cdot T_s$$

$$I_o = (1-D) \cdot I_{L,avg} \rightarrow I_{L,avg} = \frac{I_o}{(1-D)}$$

$$I_{L,peak} = I_{L,avg} + \frac{\Delta I}{2}$$

$$V_o = 10V \quad V_o = 9.99V (\text{simulation})$$

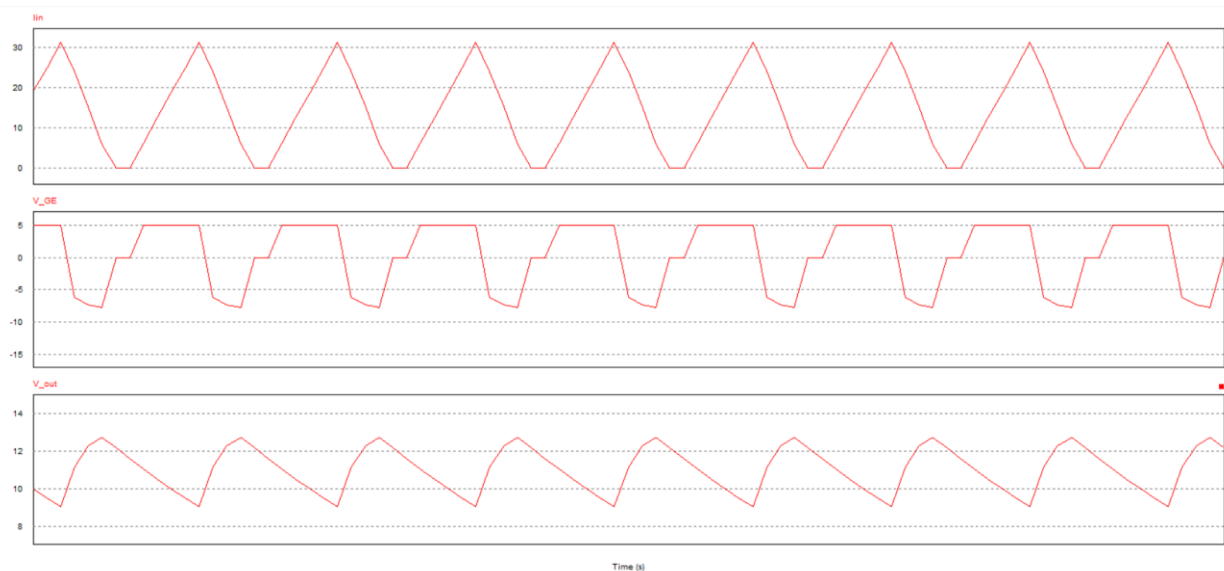
$$\frac{\Delta V_o}{V_o} = 0.268 (\text{simulation})$$

critical inductance

$$L_c = \frac{T_s \cdot V_o}{8 \cdot I_{LB}} = 6.25 \cdot 10^{-6} H$$

$$L_c = 5 \mu H \text{ was selected.}$$

$\Delta I_L = 0.52A (\text{simulation})$
 $\Delta I_L = 0.52A (\text{theoretical})$
 $I_{L,avg} = 10A (\text{theoretical})$
 $I_{L,avg} = 9.92A (\text{simulation})$
 $I_{L,peak} = 10.26A (\text{theoretical})$
 $I_{L,peak} = 10.18A (\text{simulation})$



From the current graph, the circuit is working in DCM. The input current is not continuous and becomes zero for some time, so no power is taken from the source during those intervals. This makes the current more distorted.

When the inductance is smaller, the current ripple becomes larger and the average current also increases. It can also be seen that the voltage across the inductor becomes zero when the current is zero.

The current ripple is about 25 A.

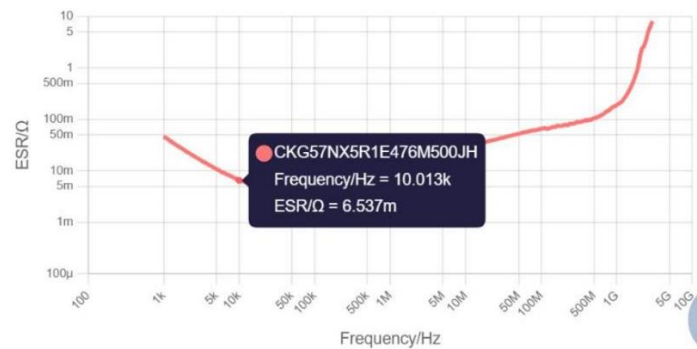
The voltage ripple is about 31%.

Components used:

Inductor: ATCA-08-251M

ATCA-08-251M	250	M	0.027	10.0	1.60
--------------	-----	---	-------	------	------

Capacitor: Multilayer Ceramic Chip Capacitor (CKG57NX5R1E476M500JH)



MOSFET: N-channel, 55 V, 0.012 Ω , 60 A,

Symbol	Parameter	Value	Unit
V_{DS}	Drain-source voltage ($V_{GS} = 0$)	55	V
V_{GS}	Gate- source voltage	± 15	V
I_D	Drain current (continuous) at $T_C = 25^\circ\text{C}$	60	A
I_D	Drain current (continuous) at $T_C = 100^\circ\text{C}$	42	A



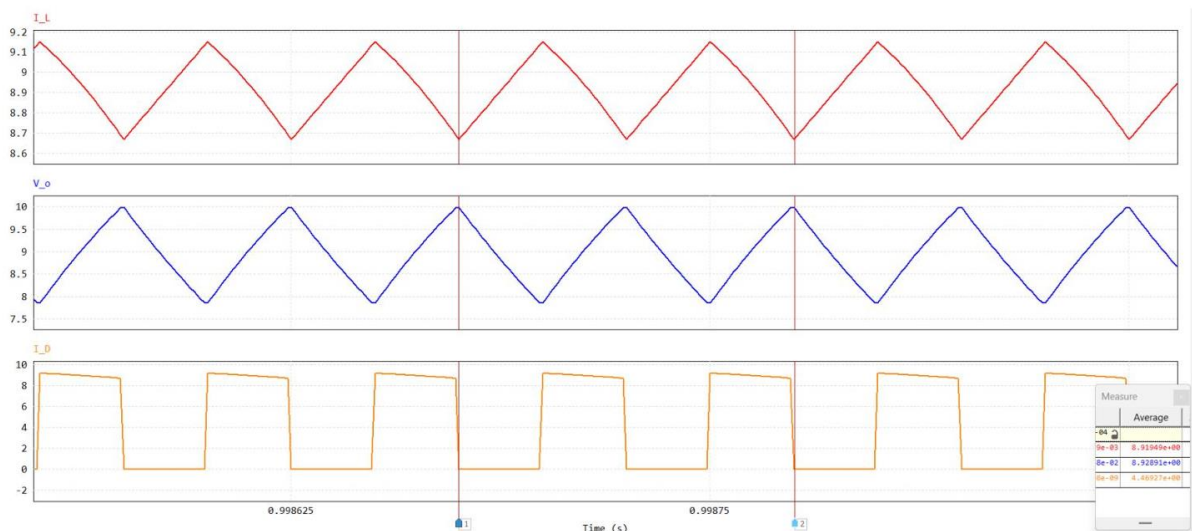
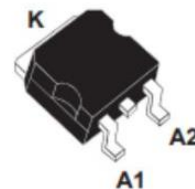
DPAK

$t_{d(on)}$	Turn-on delay time	$V_{DD} = 25V, I_D = 30A$	30	ns
t_r	Rise time	$R_G = 4.7\Omega, V_{GS} = 4.5V$	180	ns
$t_{d(off)}$	Turn-off delay time	(see Figure 12)	80	ns
t_f	Fall time		35	ns

$R_{DS(on)}$	Static drain-source on	$V_{GS} = 10V, I_D = 30A$	0.012	0.015	Ω
	resistance	$V_{GS} = 4.5V, I_D = 30A$	0.014	0.017	Ω

Schottky diode: STPS41L30CG

$I_F(\text{AV})$	2 x 20 A
V_{RRM}	30 V
$T_j(\text{max})$	150 $^\circ\text{C}$
$V_F(\text{max})$	0.38 V



The graphs of the converter after including parasitic elements and voltage drops across the components are shown.

The shape of the waveforms remains the same, but the average values change due to voltage drops on resistances and the diode, which also affects the ripple values.

The following section presents the calculation of losses and the efficiency of the converter.

Mosfet losses

conduction: $P_{C,loss} = I_o^2 \cdot R_{ds(on)} \cdot D = 10^2 \cdot 0,014 \cdot 0,5 = 0,7 W$

switching loss: $t_{on} = 30 ns$ $t_{off} = 20 ns$

$$P_s = \frac{1}{2} \cdot V_{ds} \cdot I_o \cdot (t_{on} + t_{off}) \cdot f_s = \frac{1}{2} \cdot 10 \cdot 10 \cdot (110 \times 10^{-9}) \cdot 20 \times 10^3 = 0,11 W$$

total mosfet loss

$$P_{M,loss} = P_C + P_s = 0,7 + 0,11 = 0,81 W$$

Inductor loss = $P_L = I_o^2 \cdot R_L = 10^2 \cdot 0,027 = 2,7 W$

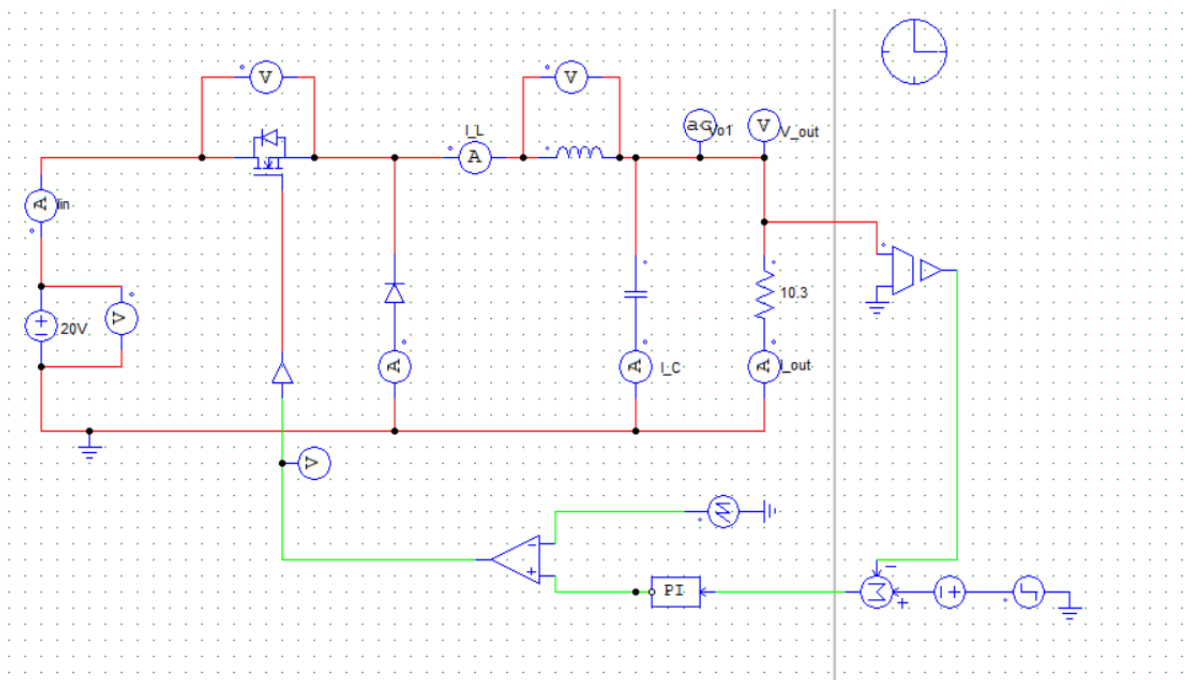
Capacitor loss = $P_{C,loss} = I_C^2 \cdot R_C \approx 0$

Diode loss = $P_D = I_o \cdot V_{dn} (1-D) = 5 \cdot 0,28 \cdot 0,5 = 0,7 W$

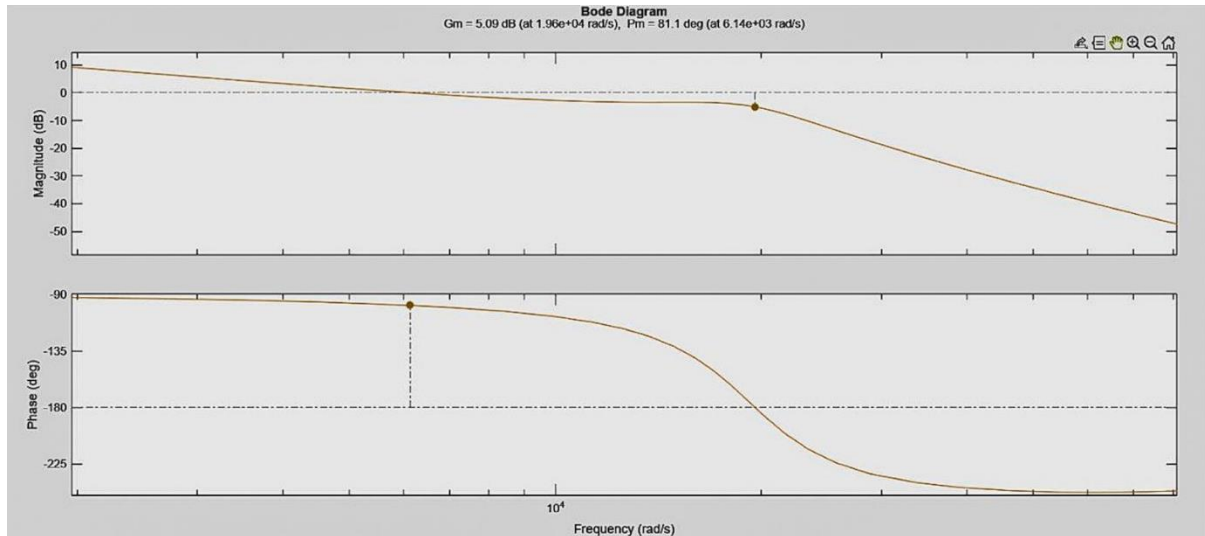
Total loss = $P_{T,loss} = P_M + P_L + P_C + P_D \approx 4,5 W$

Efficiency = $\eta = \frac{P_o}{P_o + P_{loss}} \cdot 100 = \frac{50}{54,95} \cdot 100 = 91,2 \%$

Voltage-Controlled Buck Converter:

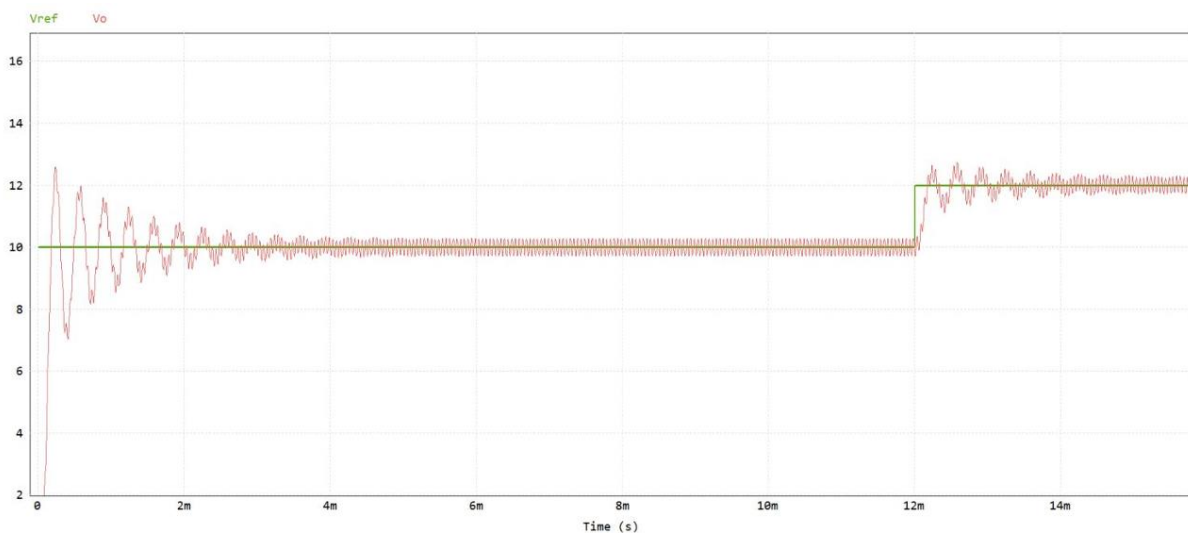


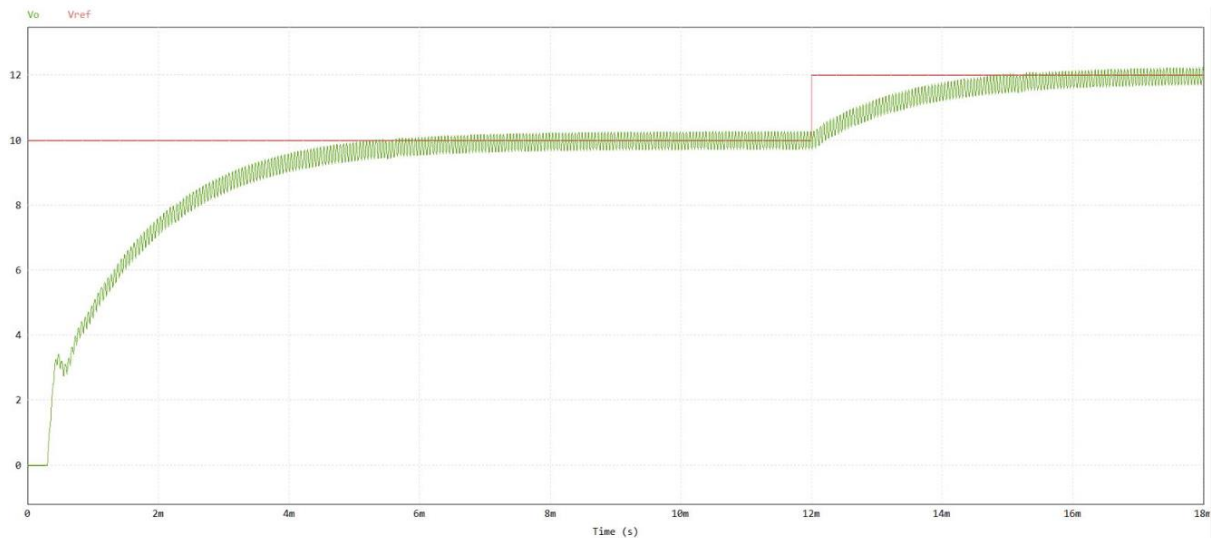
In this simulation, the buck converter with parasitic components is used, and a PI controller is added to keep the output voltage stable. It helps maintain a constant output even when the input voltage, load, or reference value changes. The PI controller also reduces the steady-state error to nearly zero while keeping the system stable.



With $K_p = 0.001$ and $K_i = 280$ The system is stable, but the trade-off is the response speed. $G_m = 5.1 \text{ dB}$ and $P_m = 81^\circ$ were obtained using these values of K_p and K_i .

The transient response oscillates, and a peak voltage of about 13 V is observed during the first transient.

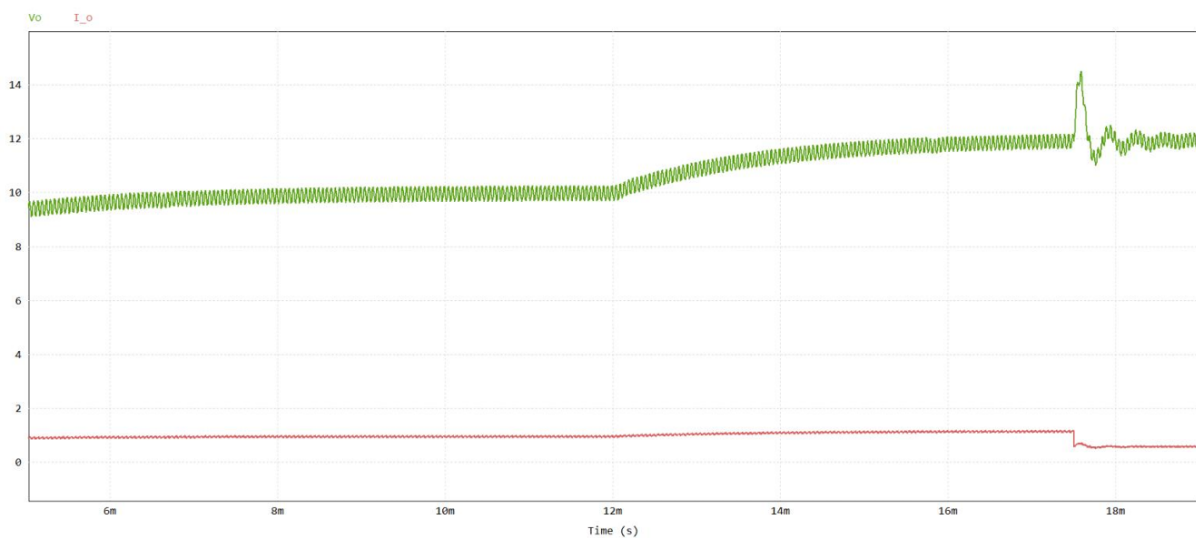




Using the final values obtained, the system shows an improved response.

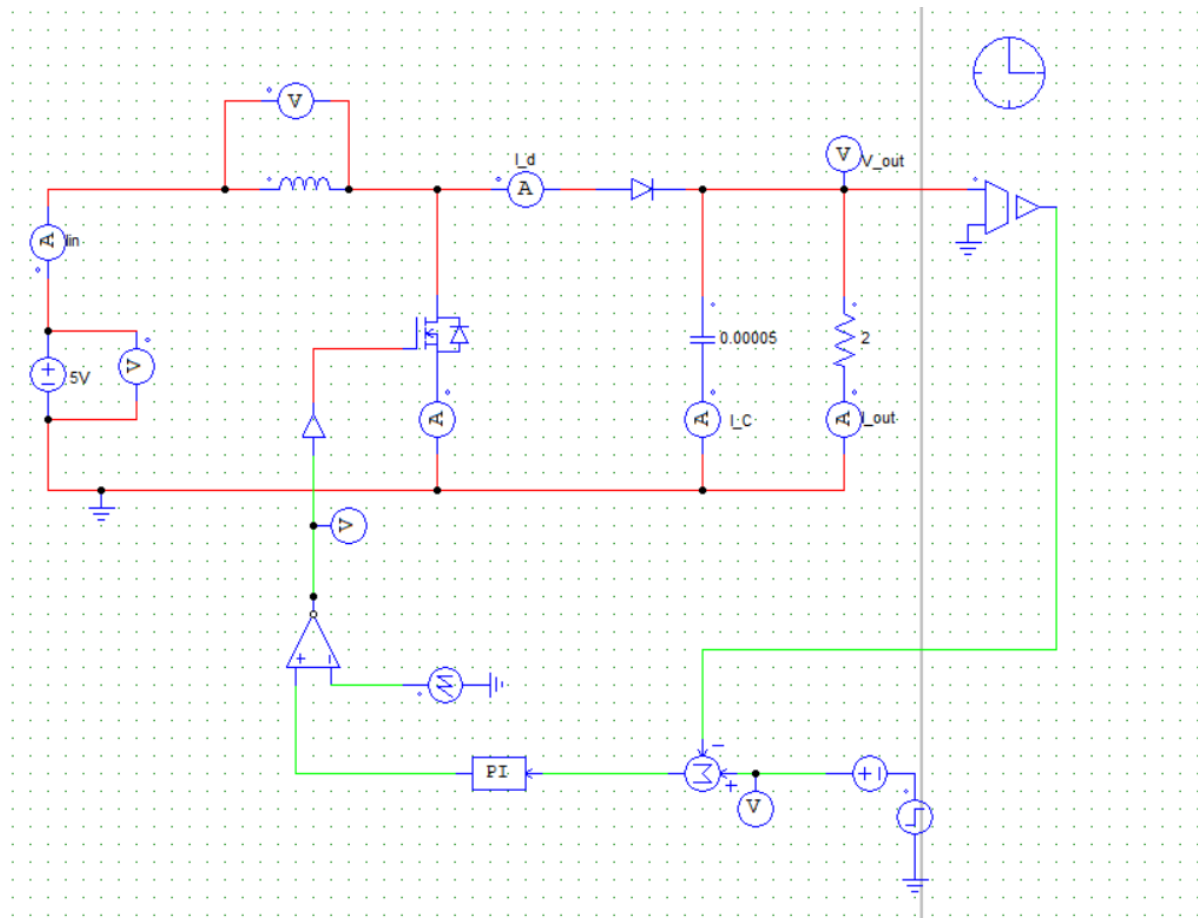
There is no voltage overshoot, the ripple remains low, and the output closely follows the reference voltage.

When the reference voltage changes from $V_{ref} = 10\text{ V}$ to $V_{ref} = 12\text{ V}$, the system takes

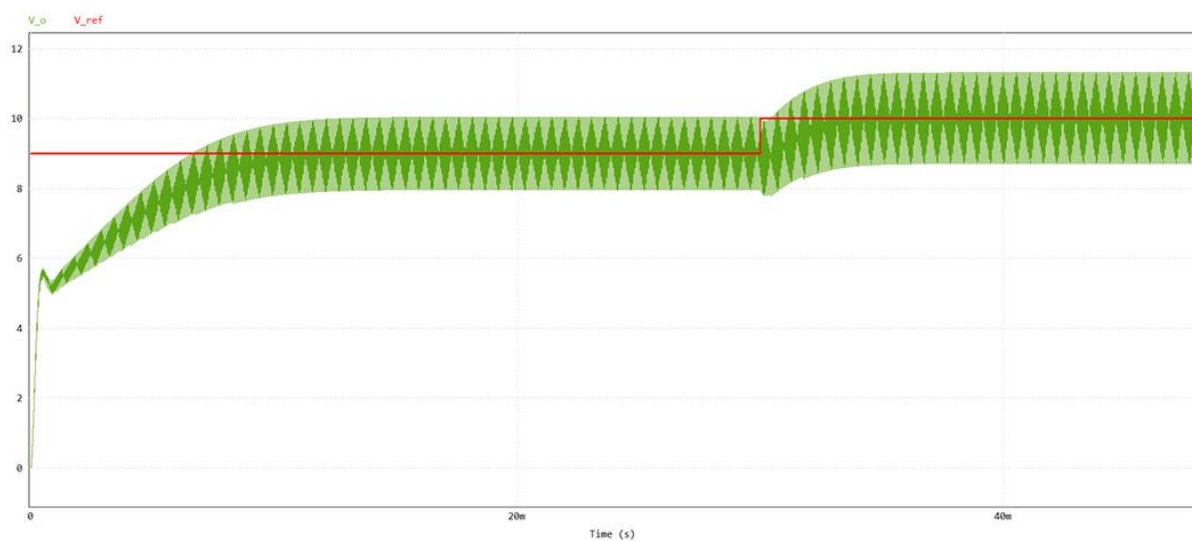


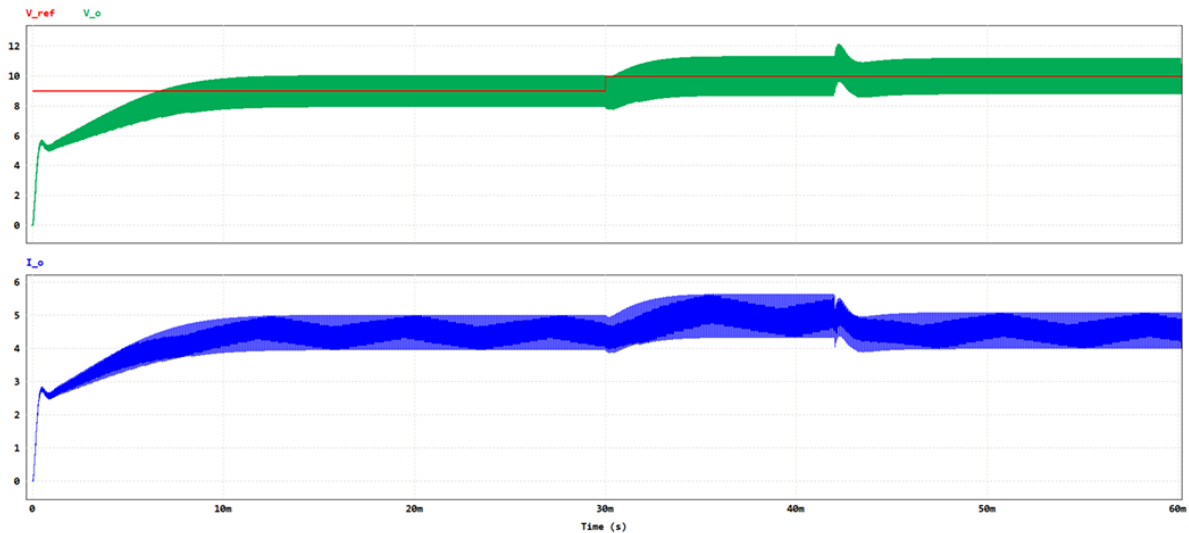
In summary, a PI controller was added with an initial reference voltage of 10 V. Then, the reference voltage was increased to 12 V, and the circuit showed a good response to this change. Once steady state was reached, a $1\ \Omega$ series resistance was added while keeping V_{ref} at 12 V. The circuit maintained an output voltage of 12 V, but some oscillations appeared when the load changed.

Voltage-Controlled Boost Converter



The following graph shows the behavior of the closed-loop circuit with a reference voltage of 9 V, which is then increased to 10 V. From this graph, it can be observed that the circuit responds well to changes in the reference voltage.

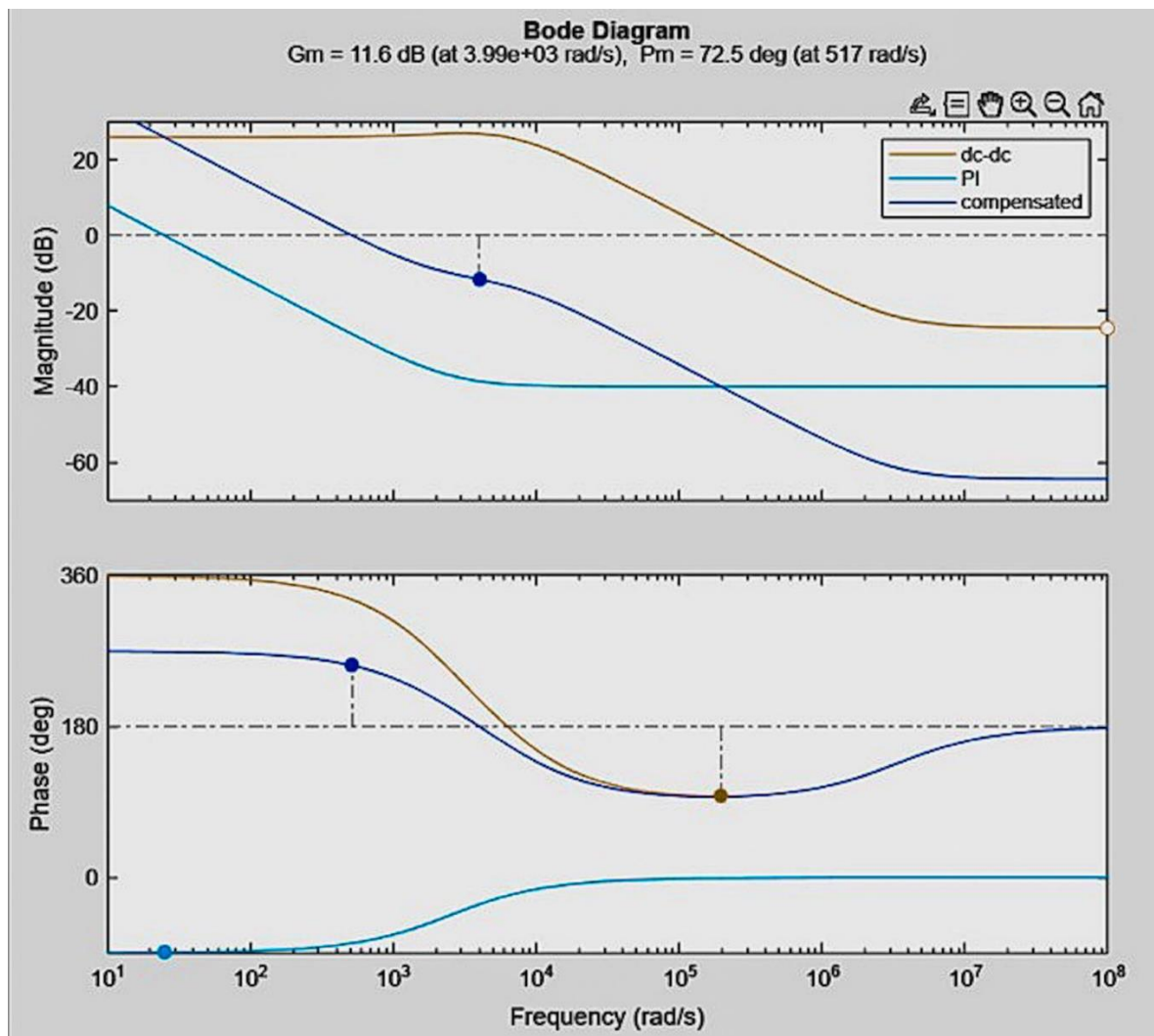




In green, the output voltage shows a disturbance caused by the switching frequency, with a ripple of about 23%, lower than in the ideal case. The reference voltage changes in a step, and the circuit does not oscillate, showing good stability.

In the next step, a resistance $R = 0.2 \Omega$ is added after the system has reached steady state.

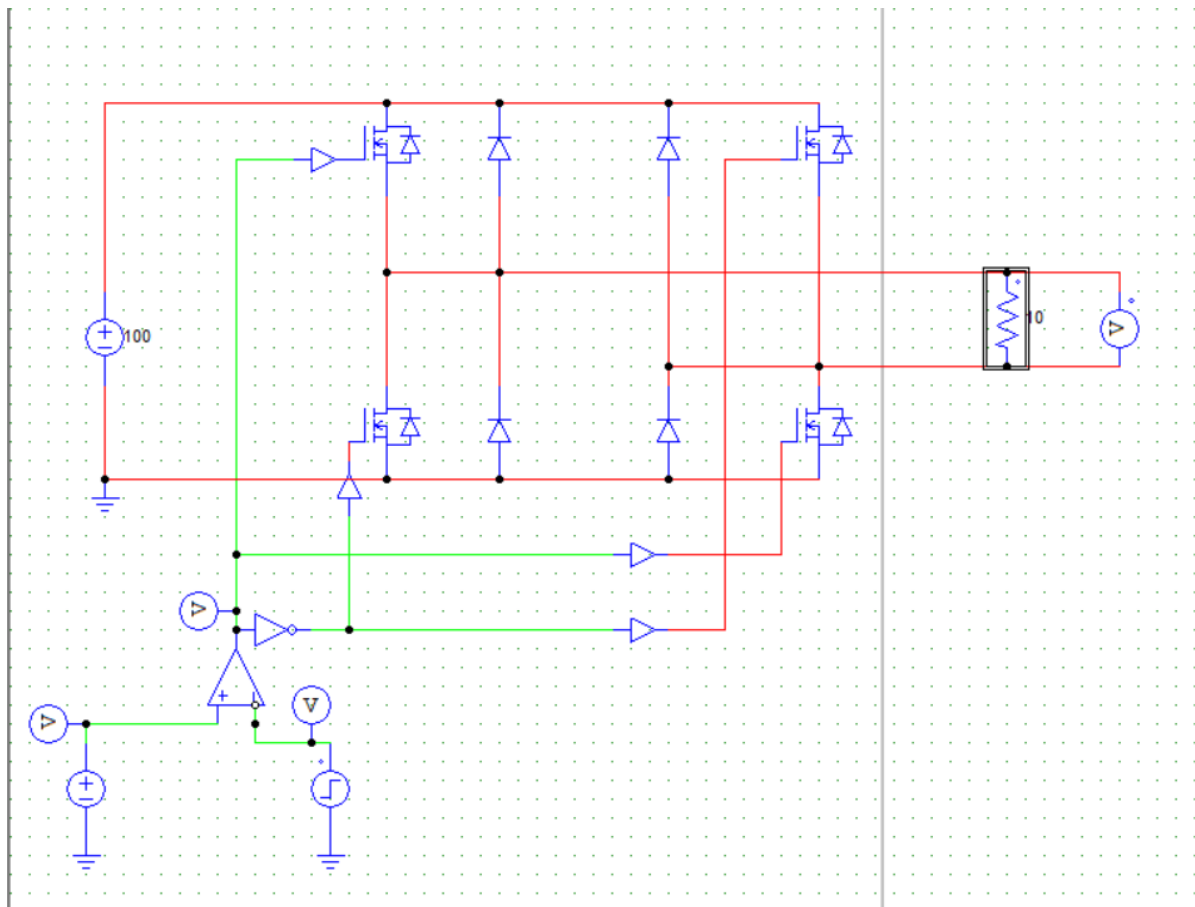
When the load changes, a small oscillation appears in the output voltage, but the controller brings it back to the reference value. The current also shows a temporary oscillation and then becomes stable again. The transient behavior is characteristic of a second-order system.



The PI controller has a pole at the origin due to the integrator (adding a 90-degree phase shift) and a zero at T around 2500 rad/s. The converter has a bandwidth of about 500 rad/s, where the phase margin is $Pm = 72.5^\circ$, and the gain margin is approximately 12 dB.

The zero of the PI improves stability against variations, while the pole increases the DC gain, reducing the steady-state error (ideally to zero). The trade-off is a slower system response.

Full-Bridge with Bipolar Switching



The full bridge circuit consists of four power switches and four diodes. The diodes allow current to continue flowing when the load is inductive.

The output voltage is taken from the middle points of the two legs, and it can switch between $+V_{in}$ and $-V_{in}$.

The switches work in pairs: when one pair is ON, the other is OFF. This makes the control simple.

The output voltage depends on the PWM control signal, so the average output voltage can be adjusted by changing the duty cycle.

To ensure that the switches operate in a complementary way, an inverter is added. The input voltage is $V_{in} = 100\text{ V}$ and the load is $R = 10\ \Omega$. The objective is to evaluate the output voltage for different duty cycle values: 0.4, 0.5, and 0.6.

in the case of a PWM modulator with output voltage and triangular wave, the duty cycle can be written as:

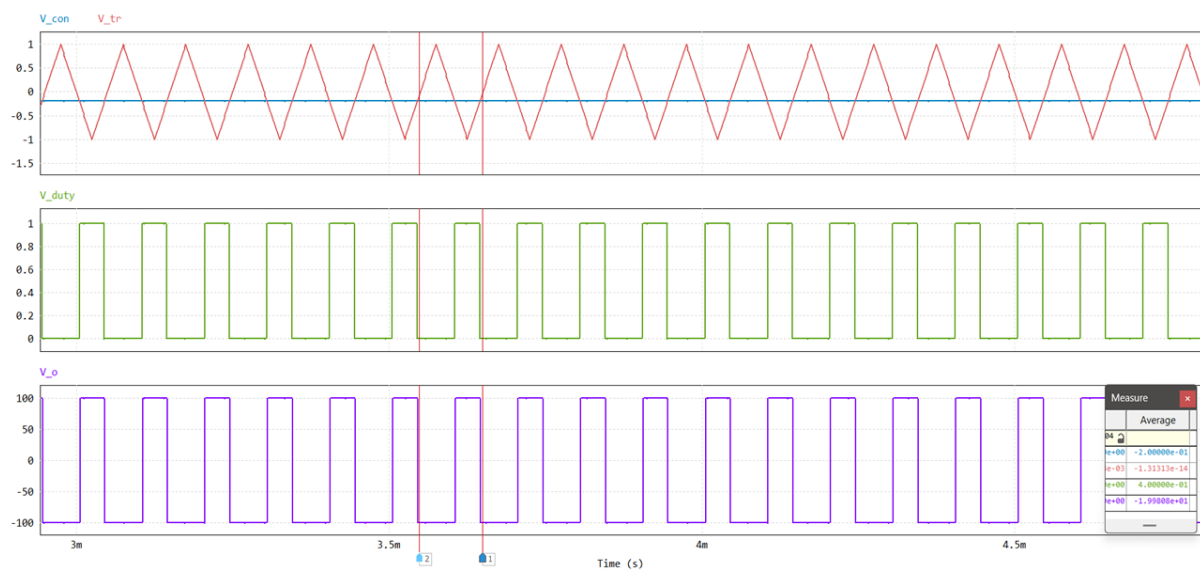
$$D = \frac{1}{2} \left(1 + \frac{V_c}{V_{tri}} \right)$$

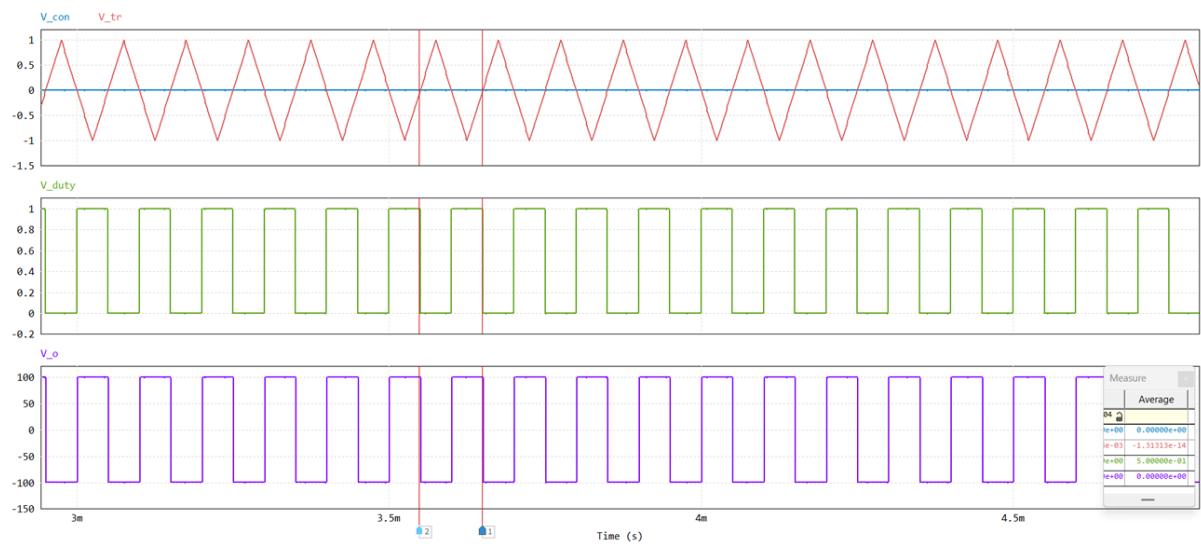
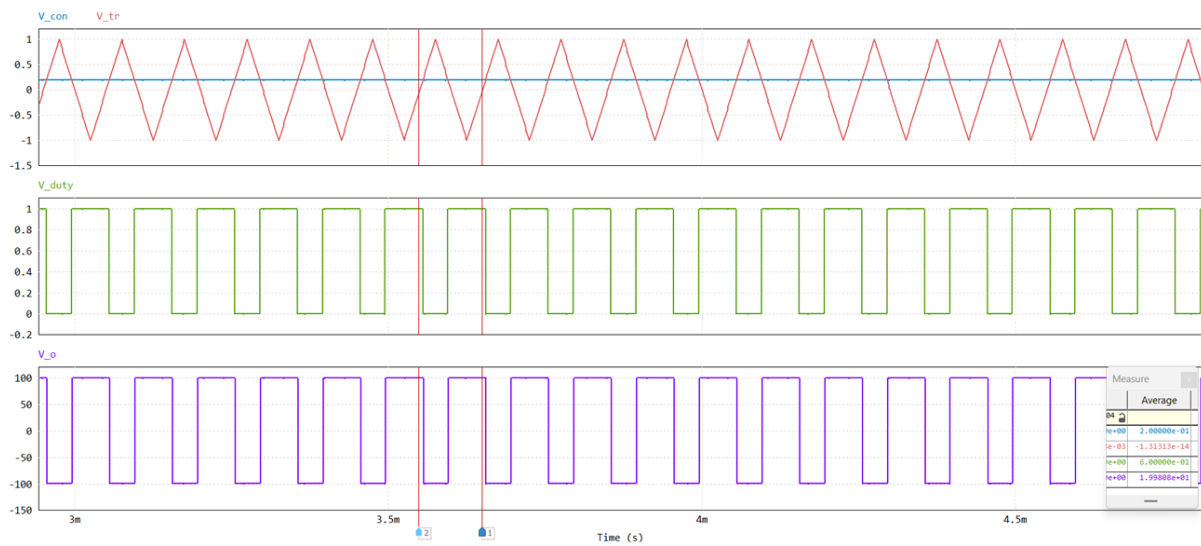
for $D = 0,4 \rightarrow 0,4 = \frac{1}{2} \left(1 + \frac{V_c}{V_{tri}} \right) = \frac{1}{2} + \frac{V_c}{2} \quad V_c = -0,2V$
 $V_{01} = 100,6(0,2) = -20V$

for $D = 0,5 \rightarrow 0,5 = \frac{1}{2} \left(1 + \frac{V_c}{V_{tri}} \right) = \frac{1}{2} + \frac{V_c}{2} \quad V_c = 0V$
 $V_{02} = 100,0 = 0V$

for $D = 0,6 \rightarrow 0,6 = \frac{1}{2} \left(1 + \frac{V_c}{V_{tri}} \right) = \frac{1}{2} + \frac{V_c}{2} \rightarrow V_c = 0,2V$
 $V_{03} = 100,02 = 20V$

Graph with $D = 0.4$. The values of V_{o_avg} are consistent with the theoretical values.

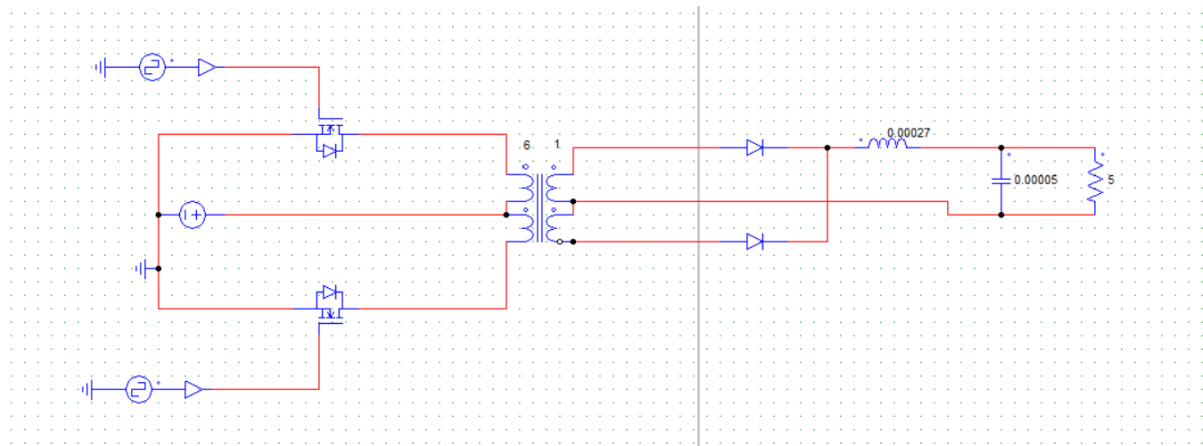


Graph with $D = 0.5$ Graph with $D=0.6$ 

Isolated DC/DC Converters

DC/DC isolated converters use a transformer. This makes the circuit more complex, but it provides important advantages. The main benefit is galvanic isolation between the input and output, which improves safety. The transformer also adds a turns ratio, allowing control of the output voltage and enabling higher voltage levels.

In this work, a push-pull converter is analyzed. It uses a transformer with a center tap on both the primary and secondary sides, two switches on the primary, and two diodes on the secondary. The diodes rectify the current and send it to an LC filter and the load. The output frequency is twice the switching frequency. The MOSFETs have their source connected to ground, which makes control easier.



I choose to operate with $D = 0.3$, from which the transformer turns ratio is obtained using the static gain of the converter. The definitions of current ripple and output voltage ripple are introduced. Using the voltage ripple definition, the difference between the maximum and minimum peak is calculated, and the cutoff frequency of the LC filter is determined to achieve the desired ripple.

Once the cutoff frequency is found, an arbitrary capacitor value is selected, and the inductance is calculated. The inductance must be greater than the critical inductance.

Handwritten calculations on grid paper:

$V_g = 300V$
 $V_o = 30V$
 $R_o = 5\Omega$
 $f_s = 20kHz$

$V_o = 2n \cdot D \cdot V_g$
 $V_c < 0.3V$
 $\frac{\Delta V_o}{V_o} < 1\%$

$n = \frac{V_o}{2 \cdot D \cdot V_g} = \frac{30}{300 \cdot 2 \cdot 0.3} = 0.25$

$n = \frac{N_2}{N_1}$ $N_1 \approx 6$

ON state = $\frac{di_L}{dt} = \frac{V_{s1} - V_o}{L}$ $\Delta I_L = \frac{V_{s1} - V_o}{L} \cdot D \cdot T_s$

Off state = $\frac{di_L}{dt} = -\frac{V_o}{L}$

$V_{o1} = n \cdot V_g$ $V_o = 2 \cdot D \cdot V_g \cdot n$ $\Delta I_L = \frac{n \cdot V_g \cdot D (1-D)}{L \cdot f_s}$

$\Delta V_o = \frac{n \cdot V_g \cdot D \cdot (1-D)}{8 \cdot f_s^2 \cdot L \cdot C} = 0.25$ (When a ripple value of 1%)

$\Delta V_o = V_o \cdot 0.01$ $\Delta V_o = 30 \cdot 0.01 = 0.3V$

cutoff frequency $\rightarrow f_c = \frac{1}{2\pi\sqrt{LC}}$ $\Delta V_o = \frac{2n \cdot V_g \cdot D \cdot (1-D)}{8} \left(\frac{f_c}{f_s}\right)^2 = 0.25$

$$\Delta V_o = \frac{(2\pi)^2 \cdot \eta \cdot V_g \cdot D \cdot (1-D)}{8} \cdot \left(\frac{f_c}{f_s}\right)^2 = 0,25$$

$$f_c = \sqrt{\frac{0,25 \cdot 8}{(2\pi)^2 \cdot \eta \cdot V_g \cdot D \cdot (1-D)}} \cdot f_s = 1419 \text{ Hz}$$

The cutoff frequency of the LC filter was determined in order to obtain a ripple of 0,25 V.

$$f_c = \frac{1}{2\pi\sqrt{LC}} \quad C = 50 \mu\text{F} \quad L = 260 \text{ nH}$$

$$L = \frac{1}{(f_c \cdot 2\pi)^2 \cdot C} = 2,52 \times 10^{-6} \text{ H}$$

This inductor and capacitor values was selected.

$$\Delta I_L = \frac{\eta \cdot V_g \cdot D \cdot (1-D)}{2L \cdot f_s} = \frac{0,16 \cdot 300 \cdot 0,3 \cdot 0,7}{260 \cdot 10^{-3} \cdot 10^3} \approx 0,3 \text{ A}$$

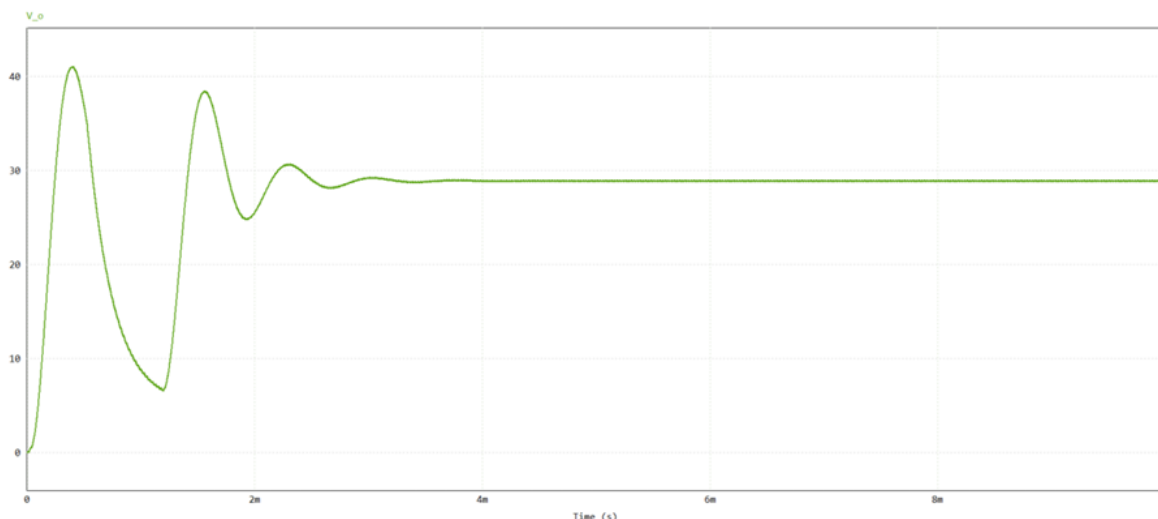
The current ripple value obtained as a result of the selected values.

Simulation results:

Current ripple = 1.11 A ; I_{L_peak} = 6.33 A

$I_o = I_{L_average}$ = 5.8 A

Voltage ripple: $V_{o_peak} - V_{o_min}$ = 0.076 V



This ripple value is much lower than required, and the output voltage shows a particular transient behavior. For this reason, keeping the inductance and ripple factor fixed, the capacitance is recalculated. A lower capacitance is obtained, which increases the ripple but improves the dynamic response of the circuit.

$$\Delta V_o = \frac{n \cdot V_g \cdot D \cdot (1-D)}{8 f_s^2 L C} = 0.25$$

$$C = \frac{n \cdot V_g \cdot D \cdot (1-D)}{8 \cdot f_s^2 \cdot L \cdot 0.25}$$

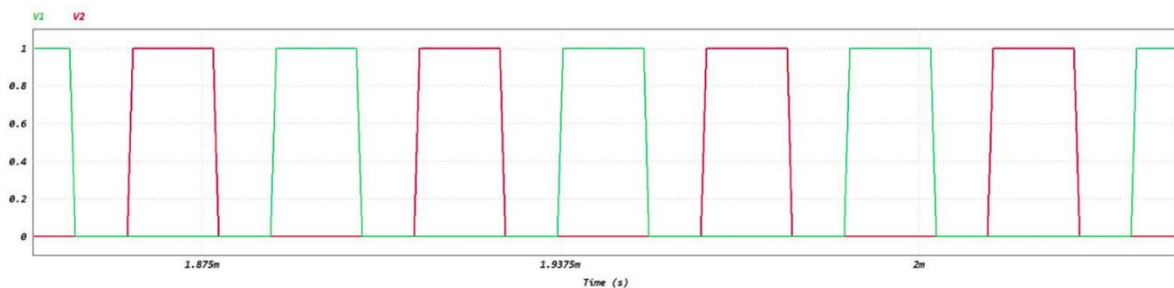
$$C = \frac{0.17 \cdot 200 \cdot 0.3 \cdot (1-0.3)}{8 \cdot (40k)^2 \cdot 260 \cdot 10^{-6} \cdot 0.25} = 1.3 \cdot 10^{-5} F$$

these values were selected and used

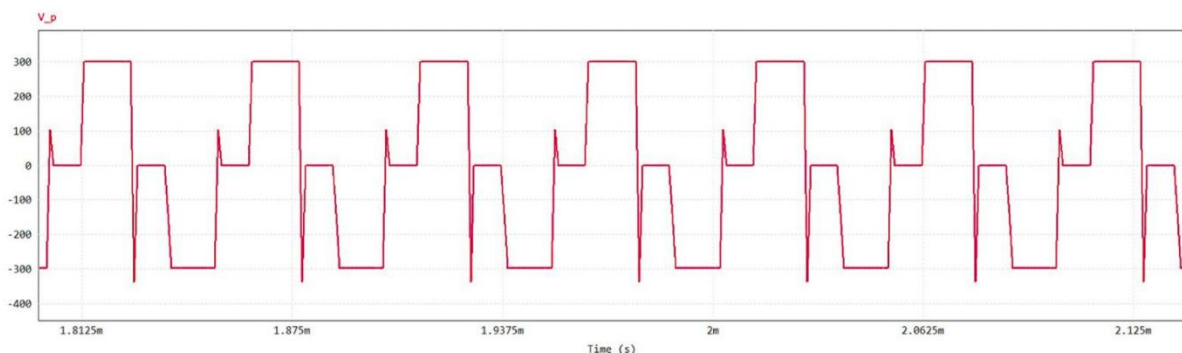
$$C = 12 \mu F \quad f_s = 40 kHz.$$

With these component values, the following waveforms are obtained:

Duty cycle signals are shifted by 180 degrees so that there is an interval where both switches are OFF.



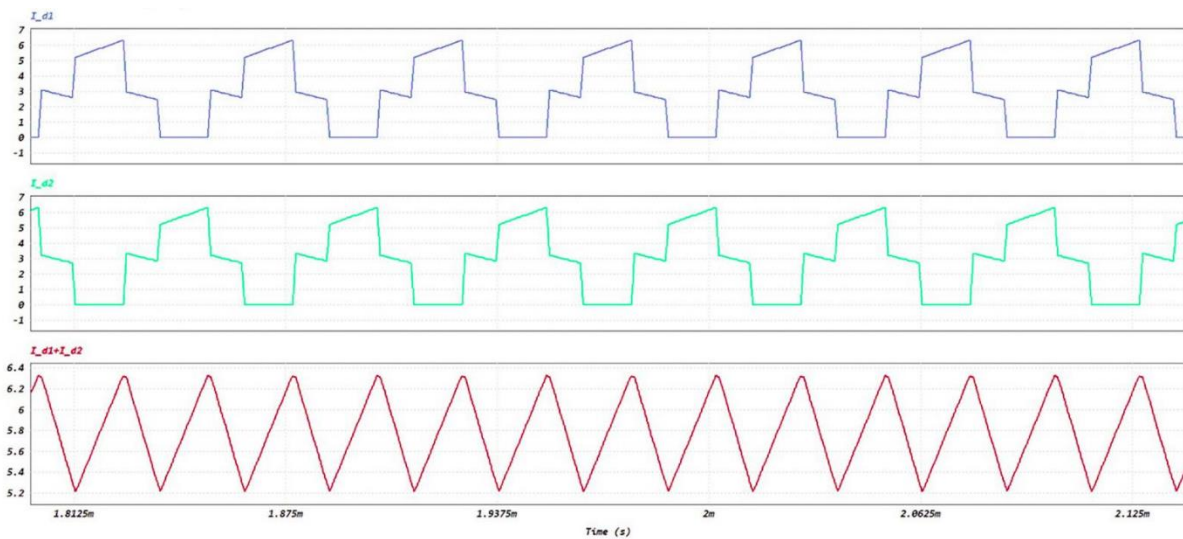
Primary voltage of the transformer: the two MOSFETs generate a waveform with two levels at $+V_g$ and $-V_g$. The sharp spikes are caused by the parasitic inductances of the transformer. A snubber circuit could be added to limit these spikes, dissipating the energy on the snubber resistor.



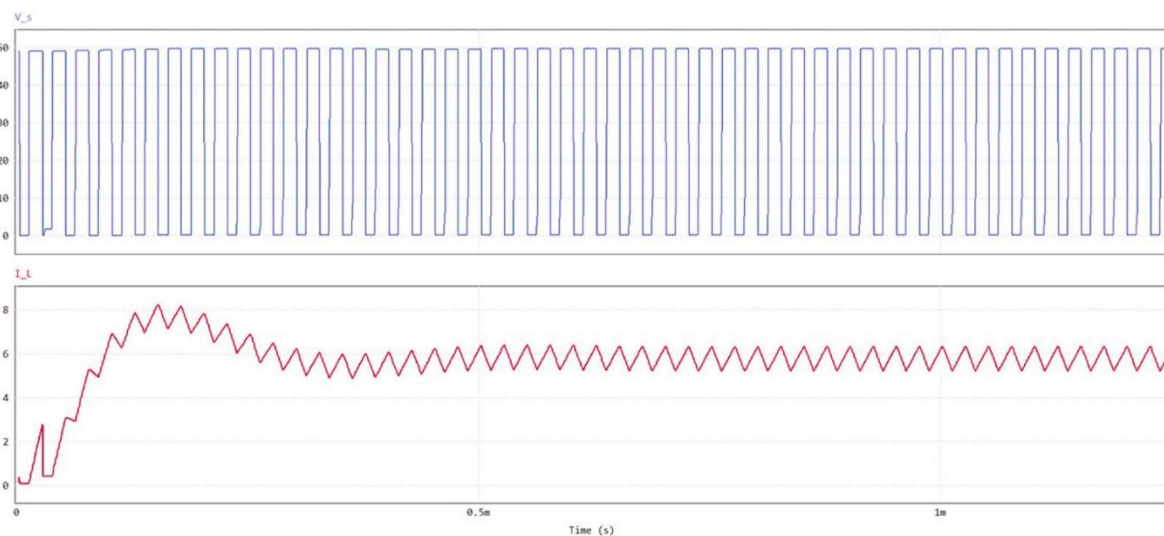
On the secondary side, the curves show the diode currents. The current is zero when the diodes are off. When Q1 is ON, D2 conducts, and when Q2 is ON, D1 conducts.

When both Q1 and Q2 are OFF, the current is about $I_o/2$. In this condition, the magnetizing inductance discharges to the secondary, and each diode carries half of the load current. If L_m is considered, each diode current also includes half of the magnetizing current .

The red curve is the sum of the two diode currents, which is equal to the output inductor current.

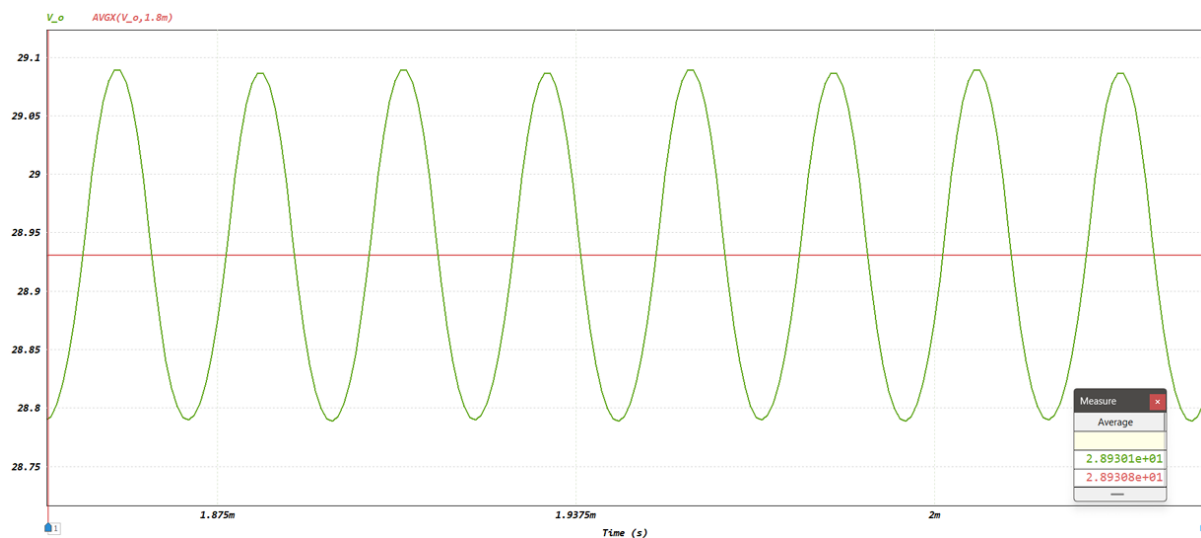
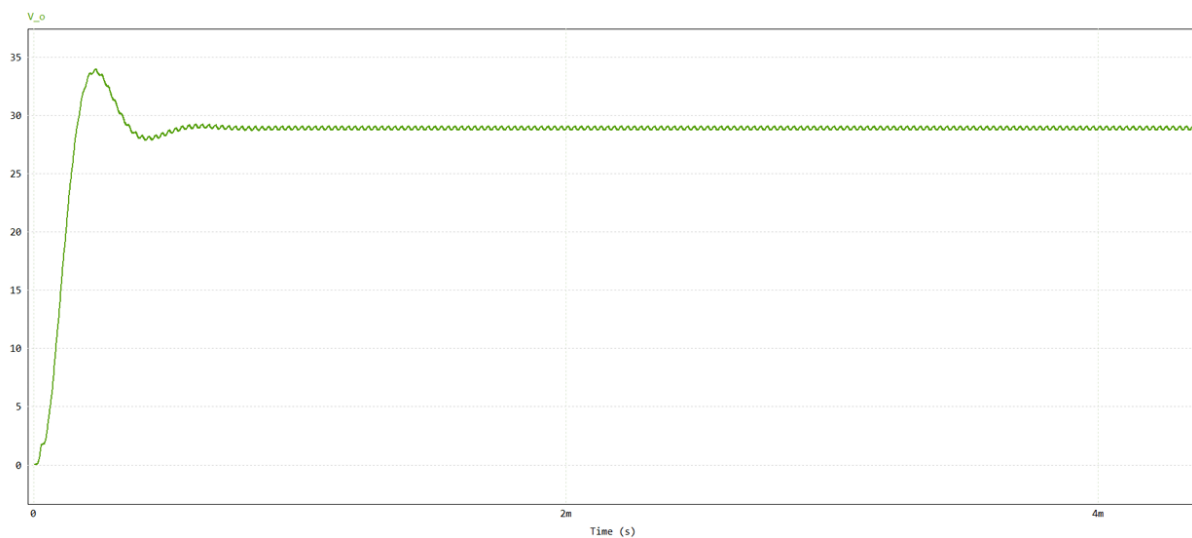


The inductor current has a triangular waveform in steady state, with a theoretical average value of 6 A. From the simulation, $I_{L_avg} = 5.8$ A, which is the current flowing to the load. The high-frequency AC component is filtered by the output capacitor.



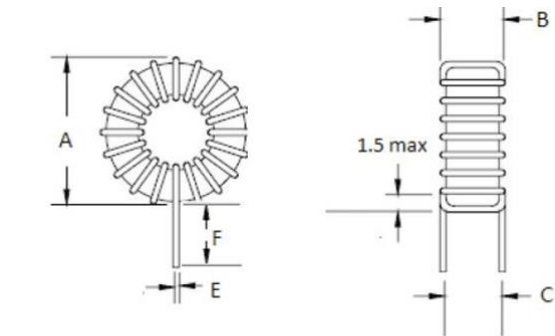
Finally, the output voltage across the load is shown. In the first graph, the transient response appears, with an underdamped second-order behavior and a peak of about 34 V. This means the load must be able to handle this voltage spike. However, the response is improved compared to the case with a larger capacitance.

In steady state, the output voltage becomes smooth, with a nearly sinusoidal shape and very low ripple. The average output voltage is about 29 V, which is close to the required 30 V.



Component selection:

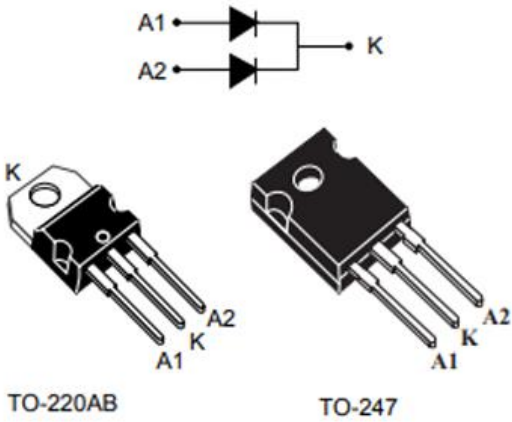
Inductor: ATCA series



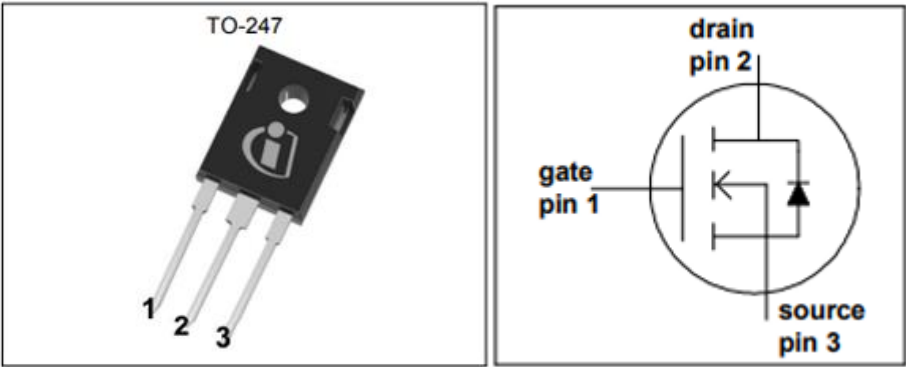
ATCA-08-251M	250	M	0.027	10.0	1.60
L			R	I_max	

Diode: STPS30H100C, Schottky diode:

Product status link	
STPS30H100C	
Product summary	
I _{F(AV)}	2 x 15 A
V _{RRM}	100 V
T _j (max)	175 °C
V _F (typ)	0.64 V



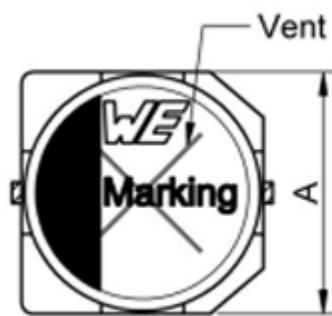
MOSFET: CoolMOS CFD2 (IPW65R080CFD)



Parameter	Value	Unit
$V_{DS} @ T_{j\ max}$	700	V
$R_{DS(on),max}$	0.08	Ω

Turn-on delay time	$t_{d(on)}$	20	ns
Rise time	t_r	18	ns
Turn-off delay time	$t_{d(off)}$	85	ns
Fall time	t_f	6	ns

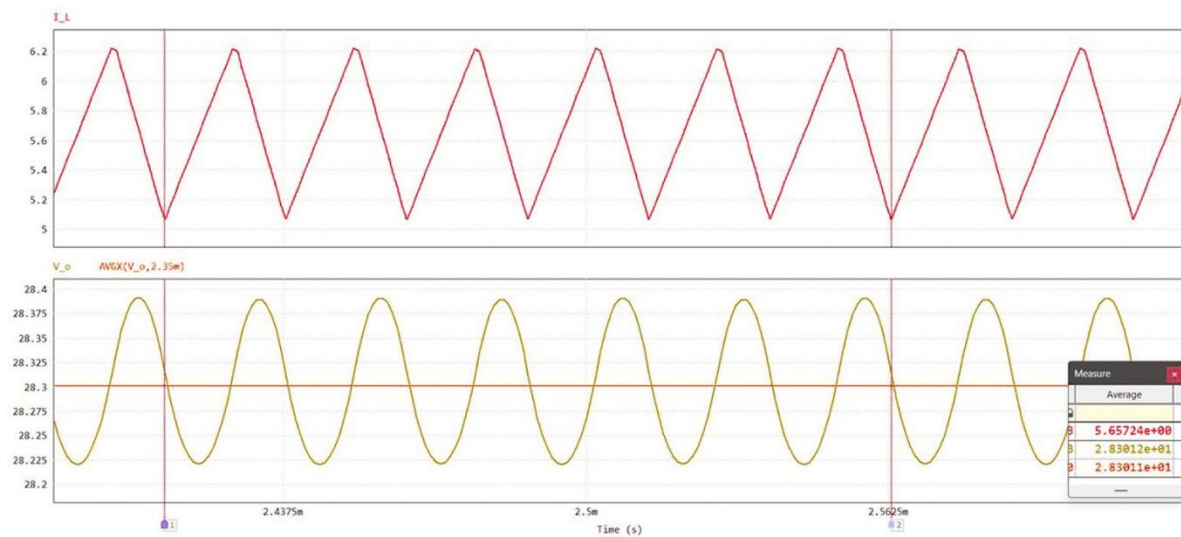
Capacitor: SMD aluminum electrolytic (WCAP-AS5H)



Properties	Test conditions		Value	Unit	Tol.
Capacitance	0.25V; 120Hz	C	0.22	μF	$\pm 20\%$
Rated voltage		U_R	50	V (DC)	max.
Leakage current	after 2 min.	I_{Leak}	3	μA	max.
Dissipation factor	120 Hz	DF	12	%	typ.
Ripple current	120 Hz @105°C	I_{ripple}	2.6	mA	max.

After including parasitic elements, the biggest impact is on the inductor current and the output voltage. The new average values are shown at the bottom right. The maximum and minimum peaks, as well as the ripple, also change.

The secondary voltage is slightly reduced compared to the ideal case, becoming $V_s = 48.7$ V instead of 49 V.



conduction losses

$$P_{c,loss} = I^2 \cdot R_{DS(on)} \cdot D \cdot T_s \cdot f_s = 1.2^2 \cdot 0.08 \cdot 0.3 = 35 \text{ mW}$$

switching losses

$$t_{on} = 20 \text{ ns}$$

$$t_{off} = 85 \text{ ns}$$

$$P_{switch} = \frac{1}{2} V_{AS} \cdot I_{in} \cdot (t_{on} + t_{off}) \cdot f_s = 300 \cdot 1.2 \cdot (105 + 105) \cdot 20 \cdot 10^3 = 0.76 \text{ W}$$

$$P_{MOSFET} = P_c + P_s = 0.8 \text{ W} \times 2 = 1.6 \text{ W}$$

capacitor losses

$$P_{g,loss} = I_E^2 \cdot R_C \approx 0$$

Diode losses

$$P_{D,loss} = 2 \cdot I_o \cdot V_{th} \cdot (1-D) = 2 \cdot 5.7 \cdot 0.64 \cdot 0.7 \approx 5.1 \text{ W}$$

Total losses

$$P_{T,loss} = P_{MOS} + P_L + P_{g,loss} + P_{D,loss} = 7.6 \text{ W}$$

Efficiency

$$\eta = \frac{P_o}{P_o + P_{T,loss}} \cdot 100 = \frac{V_o \cdot I_o}{V_o \cdot I_o + 7.6} = \frac{537.283}{283.53 + 7.6} \approx 95.5 \%$$

Current

$$I_{L,peak} = 6.21 \text{ A}$$

$$I_{L,min} = 5.07 \text{ A}$$

$$\Delta I_L = 1.14 \text{ A}$$

output voltage ripple

$$V_{o,peak} = 28.4 \text{ V}$$

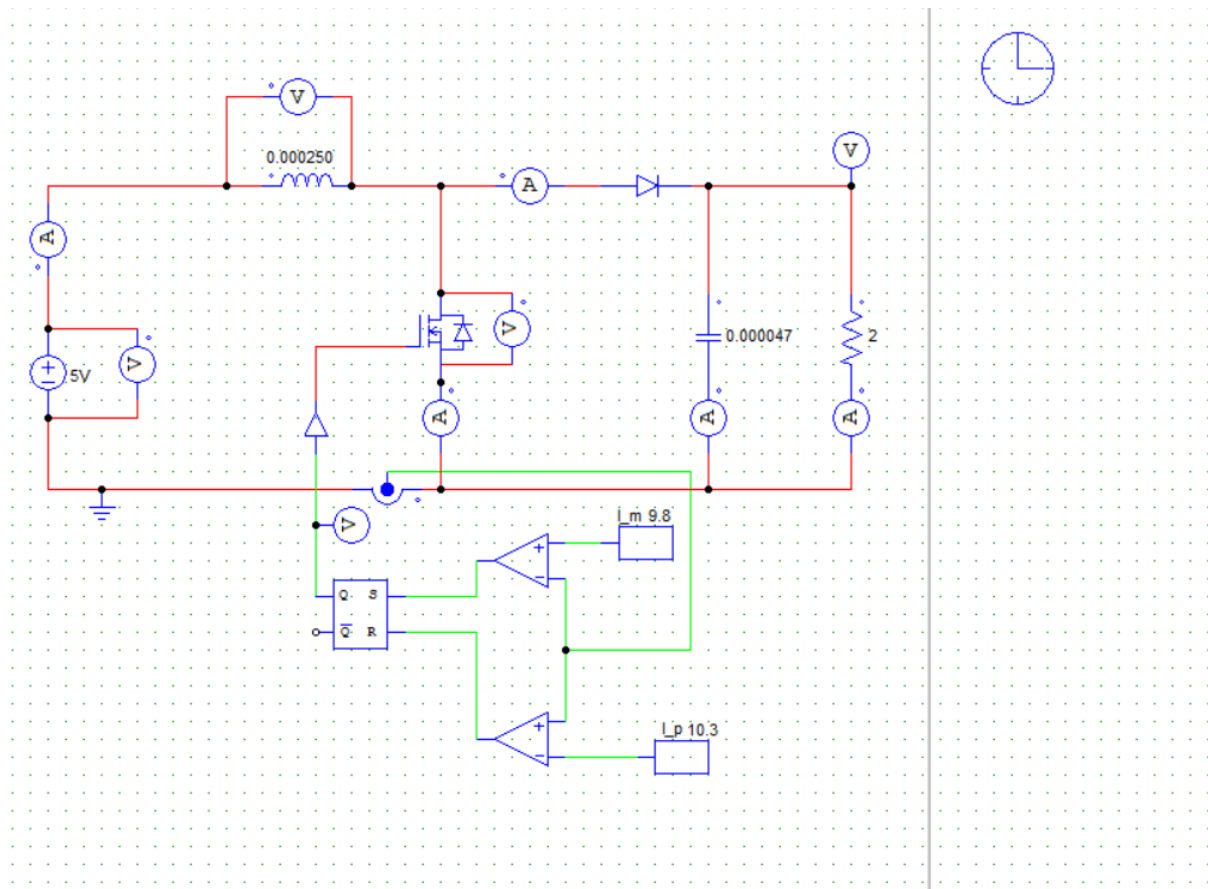
$$V_{o,min} = 28.2 \text{ V}$$

$$\Delta V_o = 0.2 \text{ V}$$

input voltage

$$V_{s,max} = 48.8 \text{ V}$$

Boost DC/DC Converter with Hysteresis Control:



In hysteresis control, the input current (in a boost converter, the inductor current) is compared with two limits: an upper and a lower threshold around the desired value.

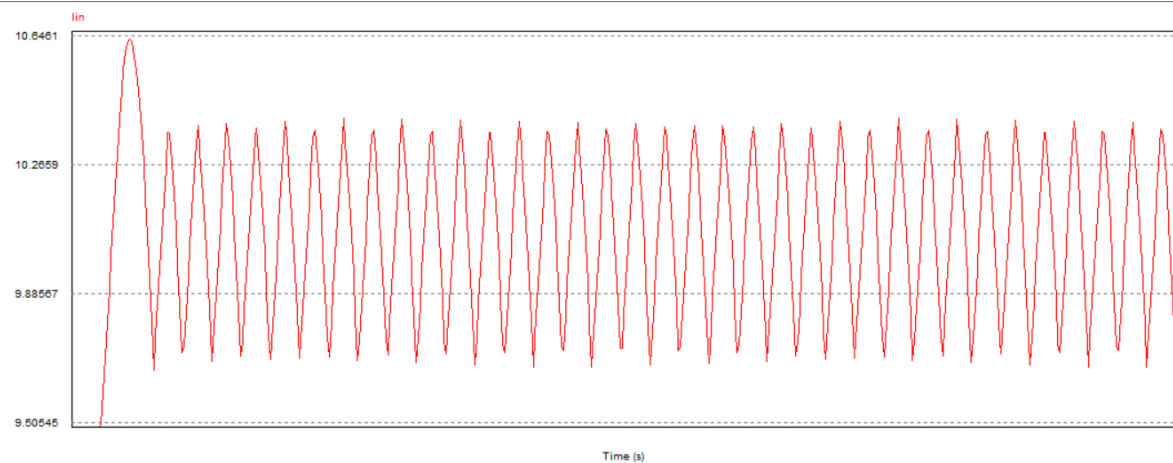
When the current goes above the upper limit, the switch is turned OFF. When the current goes below the lower limit, the switch is turned ON. In this way, the current is kept within a certain band, called the tolerance band.

The control system includes a current sensor, which is more complex than a voltage sensor. The measured current is compared with the two thresholds, and the result is sent to a set-reset flip-flop. If the current is too low, the flip-flop turns the switch ON; if it is too high, it turns the switch OFF.

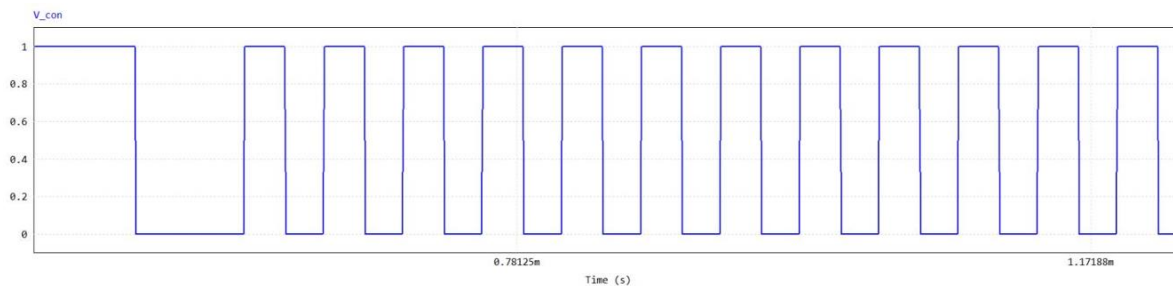
This control method changes the switching frequency instead of keeping it constant. A narrow tolerance band means small current variation, but it also leads to a higher switching frequency and higher losses.

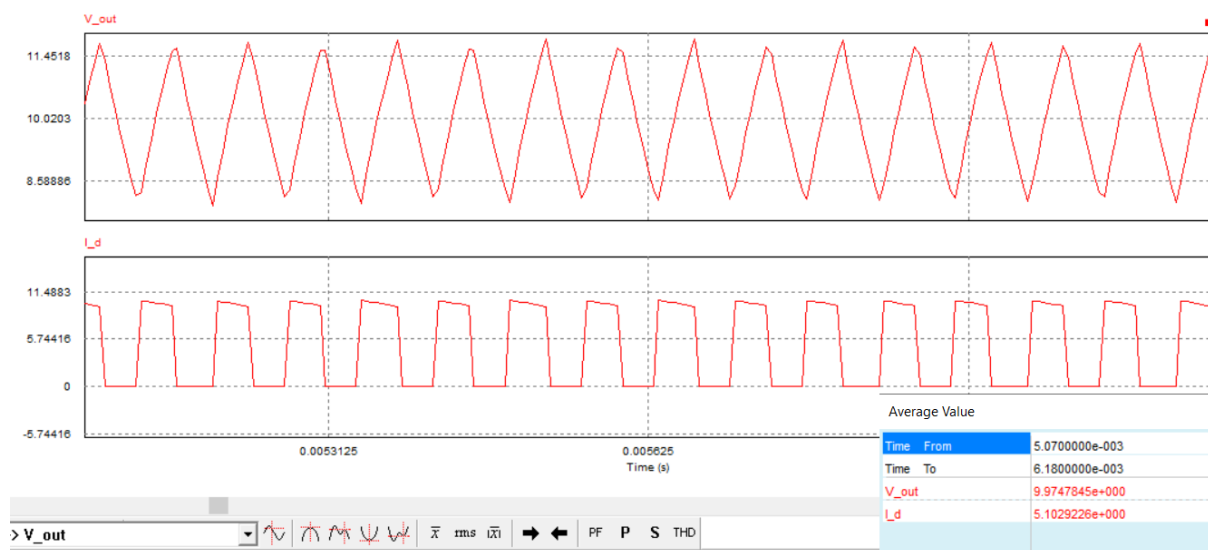
In this example, the tolerance band is set around $I_L = 10$ A to obtain an output current of $I_o = 5$ A and keep the output voltage at $V_o = 10$ V.

An important advantage of current control is that the system behaves like a first-order system, so there is no overshoot and less stress on the switch and the diode.



The graph shows the tolerance band for the inductor current and how the duty cycle changes. When the current goes above the upper limit, the switch turns OFF. In this case, the inductor sends energy to the load, and the current decreases until it reaches the lower limit. Then, the switch turns ON again, the inductor current increases, and the cycle repeats. This process keeps the current stable, and as a result, the output voltage remains close to the desired value.





From this graph, it can be observed that the diode conducts with a current equal to the inductor current when the power switch is off. During this interval, the capacitor charges and the output voltage increases.

When the switch turns on, the diode becomes reverse-biased, and the capacitor is the only element supplying energy to the load, causing its voltage to decrease.

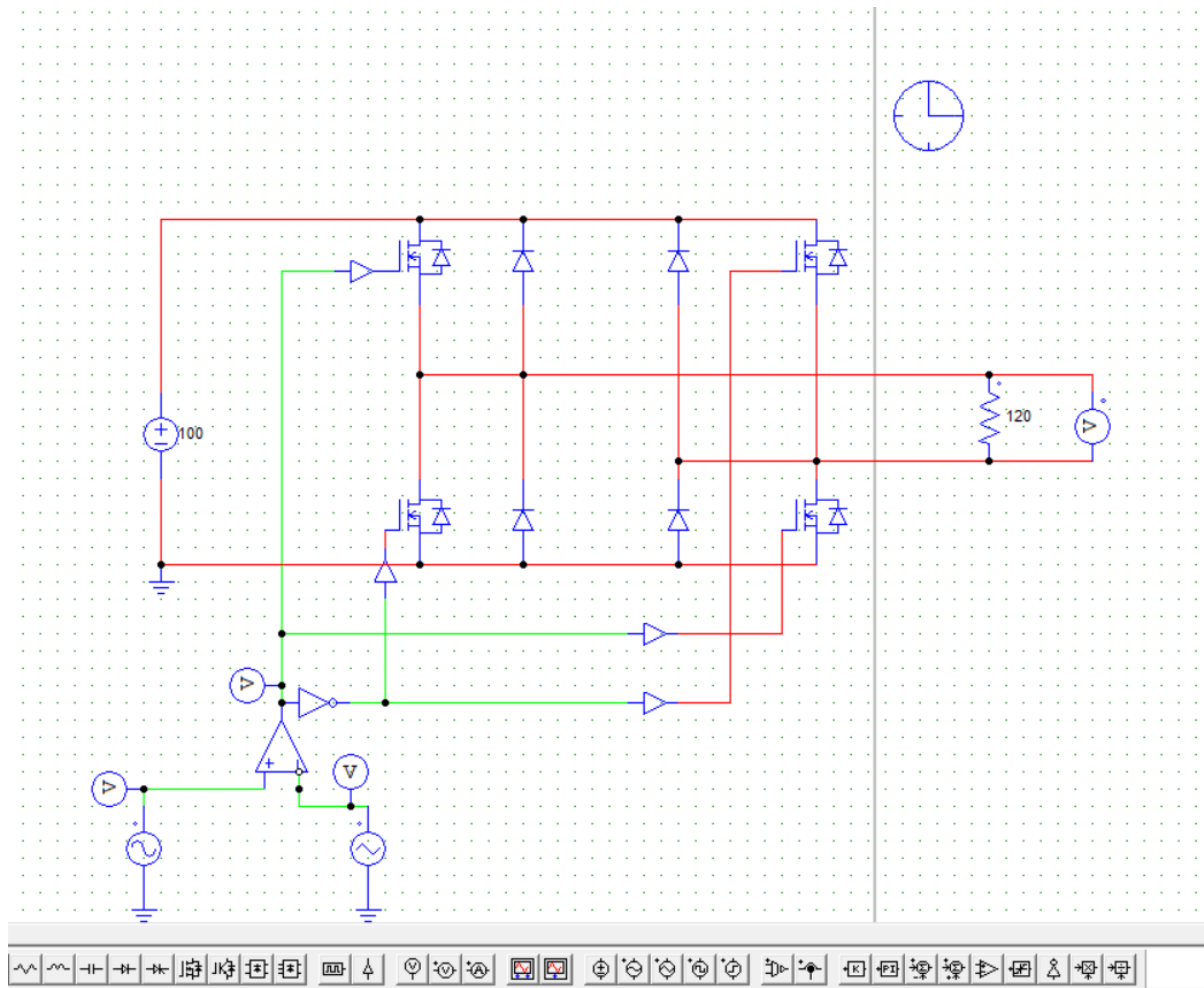
INVERTER :

An inverter is a device that converts DC into AC .In other words, it takes a constant voltage and produces a sinusoidal output.

In the simplest case, a full bridge circuit can be used to generate a square wave from a DC source. Then, by filtering this signal, the main sinusoidal component can be obtained as the output.

Unlike DC/DC converters, in inverters a sinusoidal control signal is used. This means the duty cycle changes over time in a smooth, sinusoidal way. As a result, the output voltage also follows this sinusoidal shape, producing an AC waveform.

DC/AC Inverter with Full-Bridge Bipolar Switching:

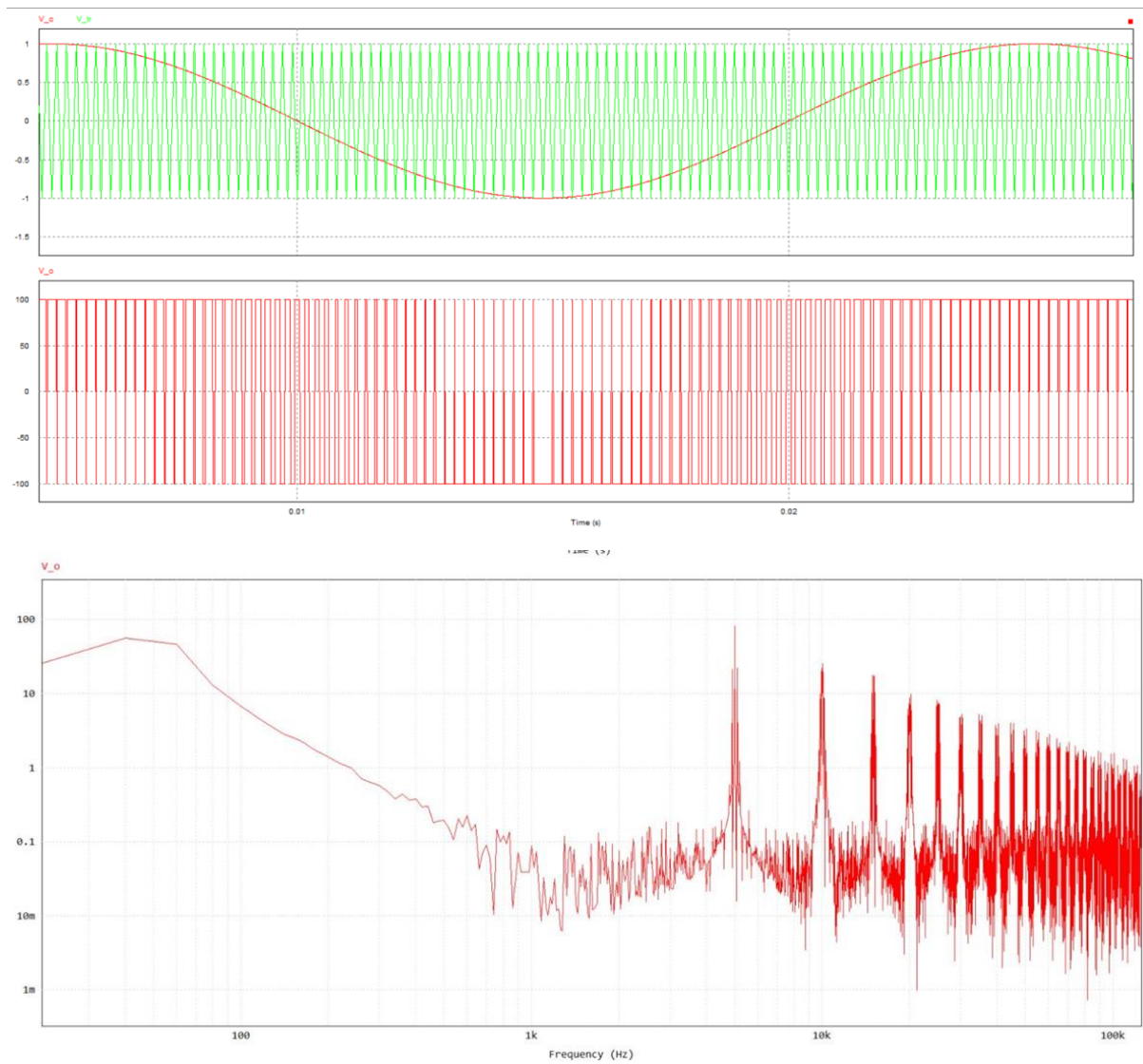


In the simulation, the modulation index is varied while keeping the frequency modulation constant to observe changes in harmonic content.

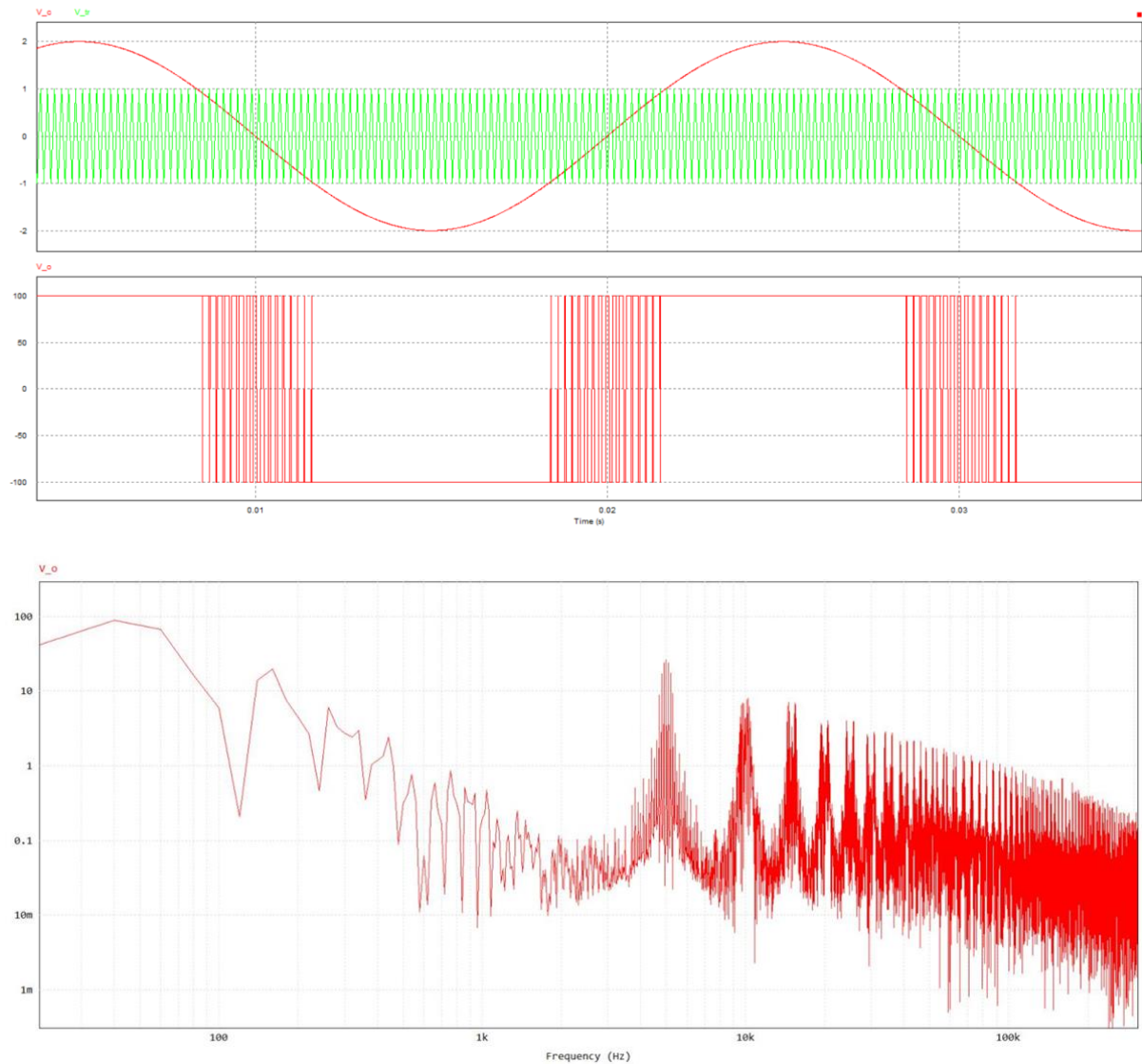
For low m_a , there are few harmonics but with high amplitudes, sometimes higher than the fundamental. As m_a increases, the number of harmonics increases but their amplitudes decrease.

When $m_a > 1$, the inverter enters overmodulation: the fundamental is no longer linear and harmonic distortion increases. In saturation, the waveform becomes square with high harmonic content.

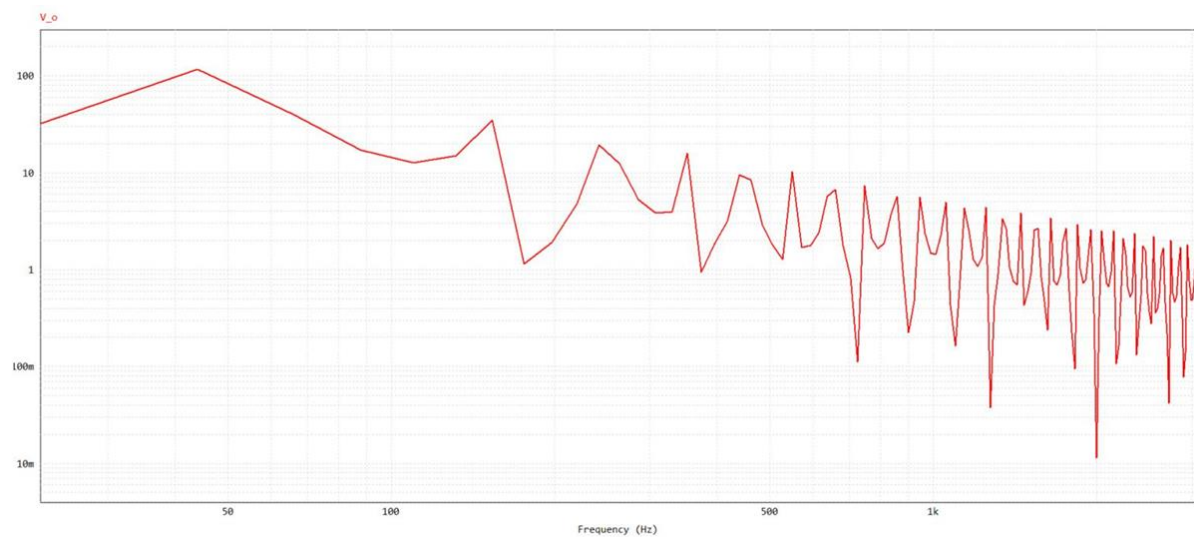
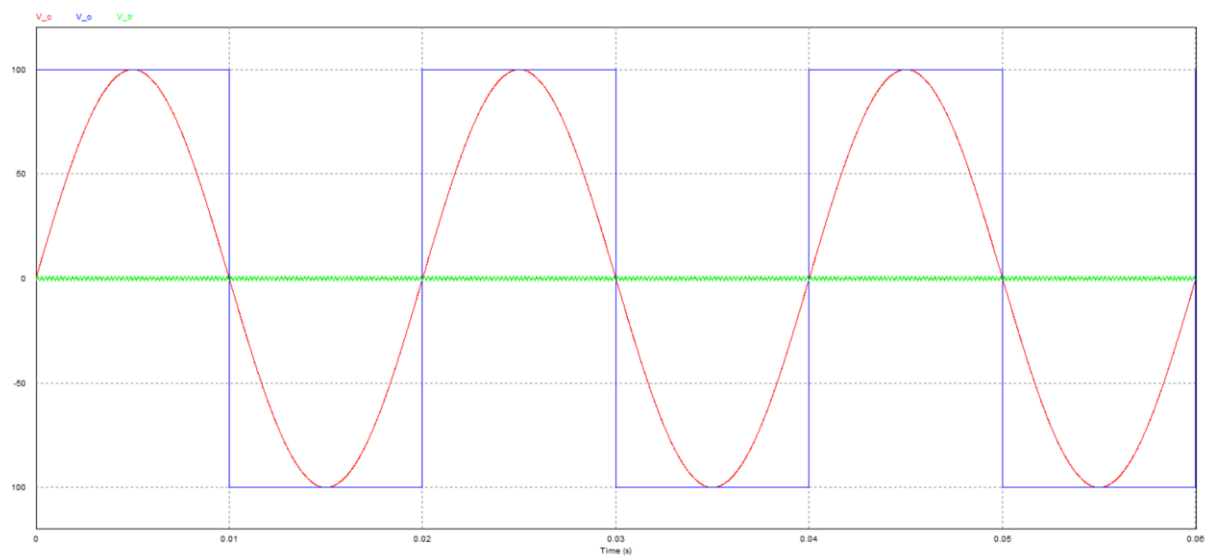
$m_f = 100$ and $m_a = 0.8$



$m_f = 100$ and $m_a = 2$



$m_f = 100$ and $m_a = 100$



$V_{in} = 100V$ $R = 120\Omega$
 $f_1 = 50Hz$ $m_f = \frac{f_s}{f_1} \rightarrow f_s = 100 \cdot 50 = 5kHz$
 $m_a = 0,8 = \frac{V_c}{V_{tri}}$ $V_{tri} = 1V \rightarrow V_c = 0,8V$
 $m_a = 2 \rightarrow V_c = 2V$
 $m_a = 100 \rightarrow V_c = 100V$

In overmodulation, the amplitude of the fundamental no longer increases linearly. Beyond a certain value of m_a , the amplitude of the fundamental saturates. At the output, a quasi-square waveform is obtained.

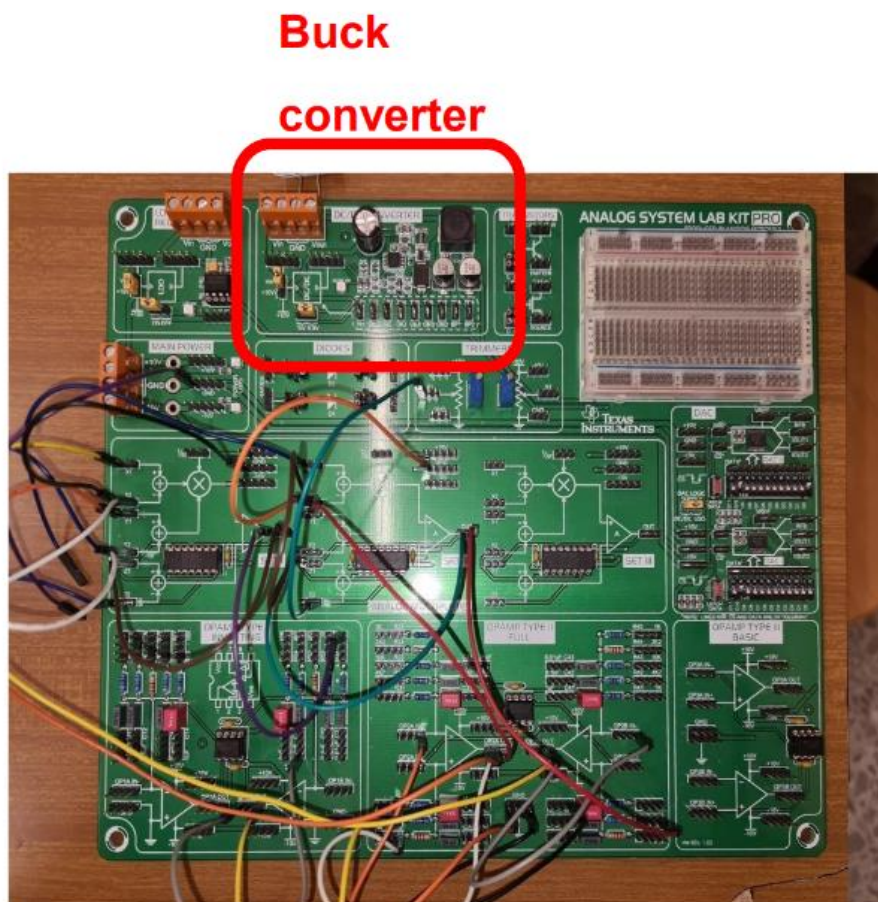
For $m_a = 0.8$, the fundamental at 50 Hz has an amplitude similar to the harmonic at the switching frequency.

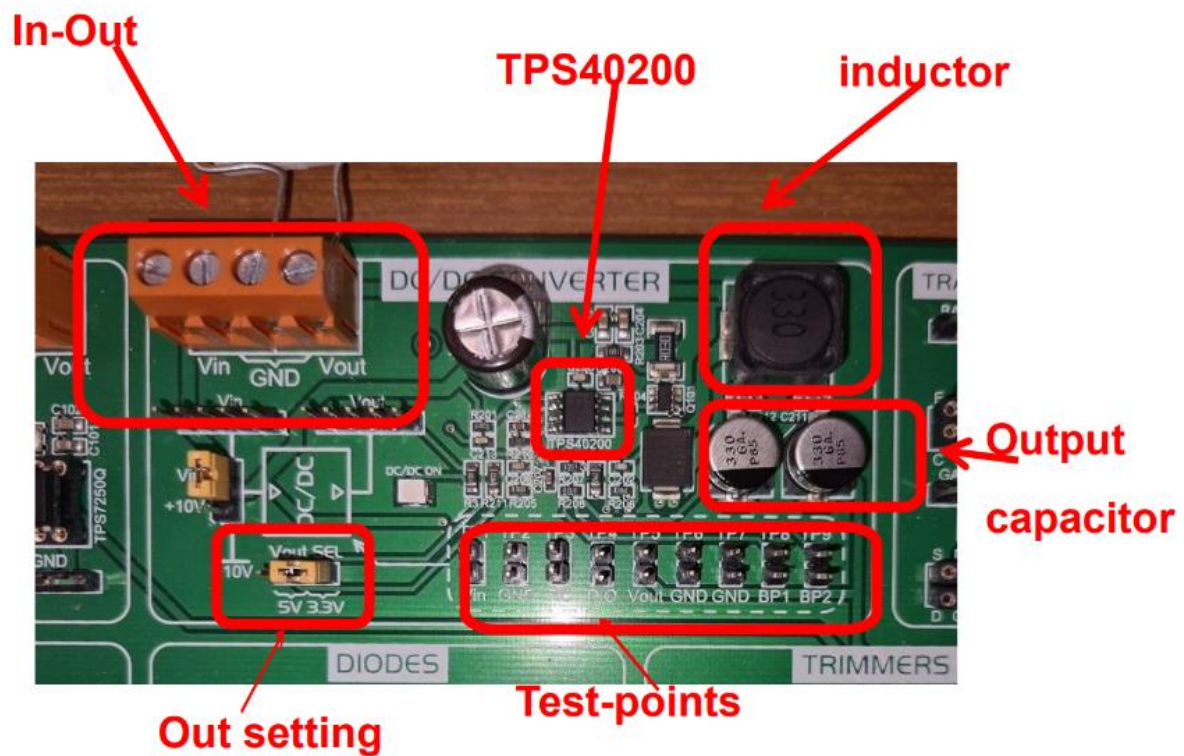
For $m_a = 2$, the control signal does not always intersect the carrier, so the switches stay on for some intervals. This makes the output more like a square wave and introduces lower-frequency harmonics.

For $m_a = 100$, the signal behaves like a square wave, with harmonics at odd multiples of 50 Hz (50, 150, 250 Hz, etc.).

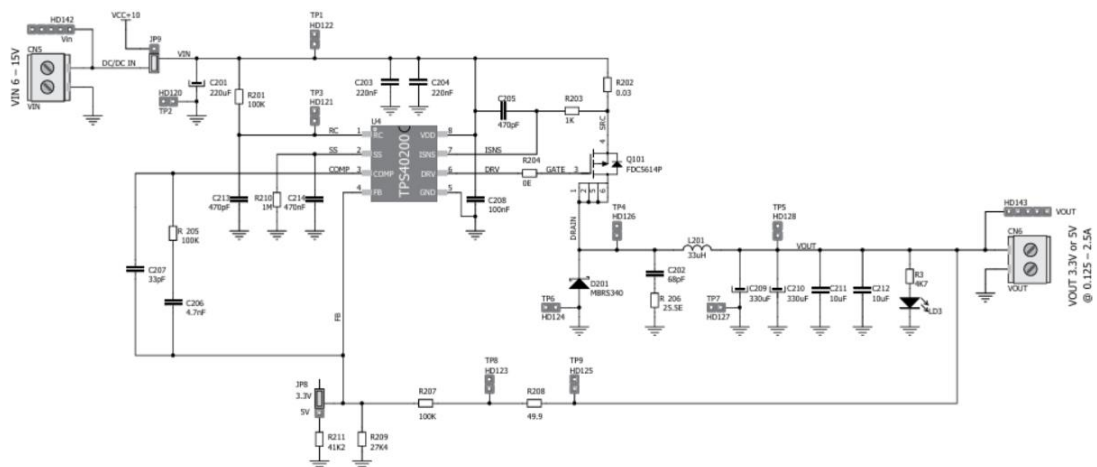
LABORATORY:

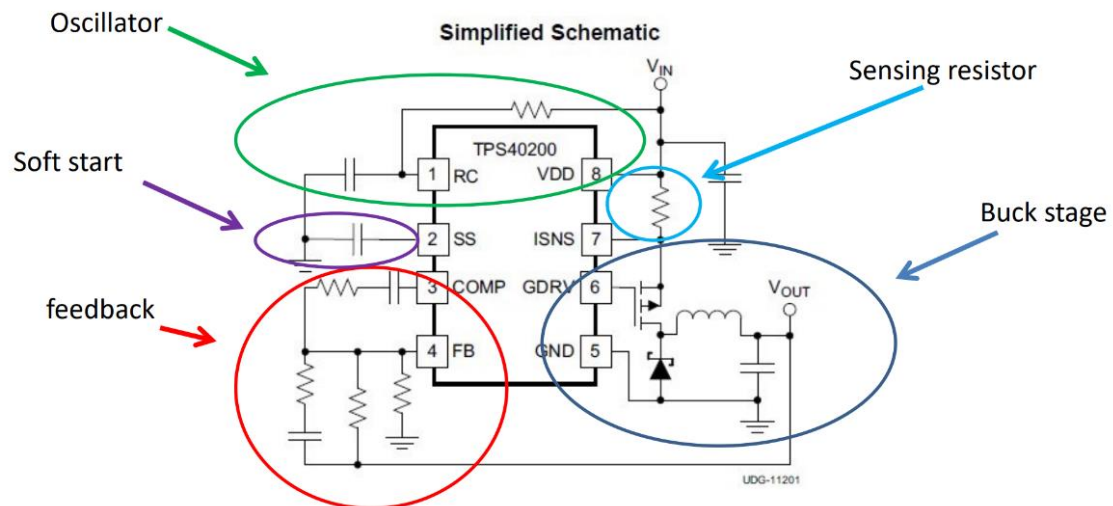
This laboratory session focuses on the experimental characterization of a DC-DC buck converter integrated with the TPS40200 controller. The objective was to analyze how the commercial converter operates and to observe its behavior in a real circuit.





Buck converter





The converter includes an input capacitor to support the supply and filter fluctuations. The switching frequency is 200 kHz, allowing small components. An oscilloscope was used to observe waveforms, and a multimeter measured parameters like the sensing resistance. The output voltage is taken through a voltage divider and sent to the comparator. A jumper allows adding a parallel resistor to adjust the feedback factor.

Procedure:

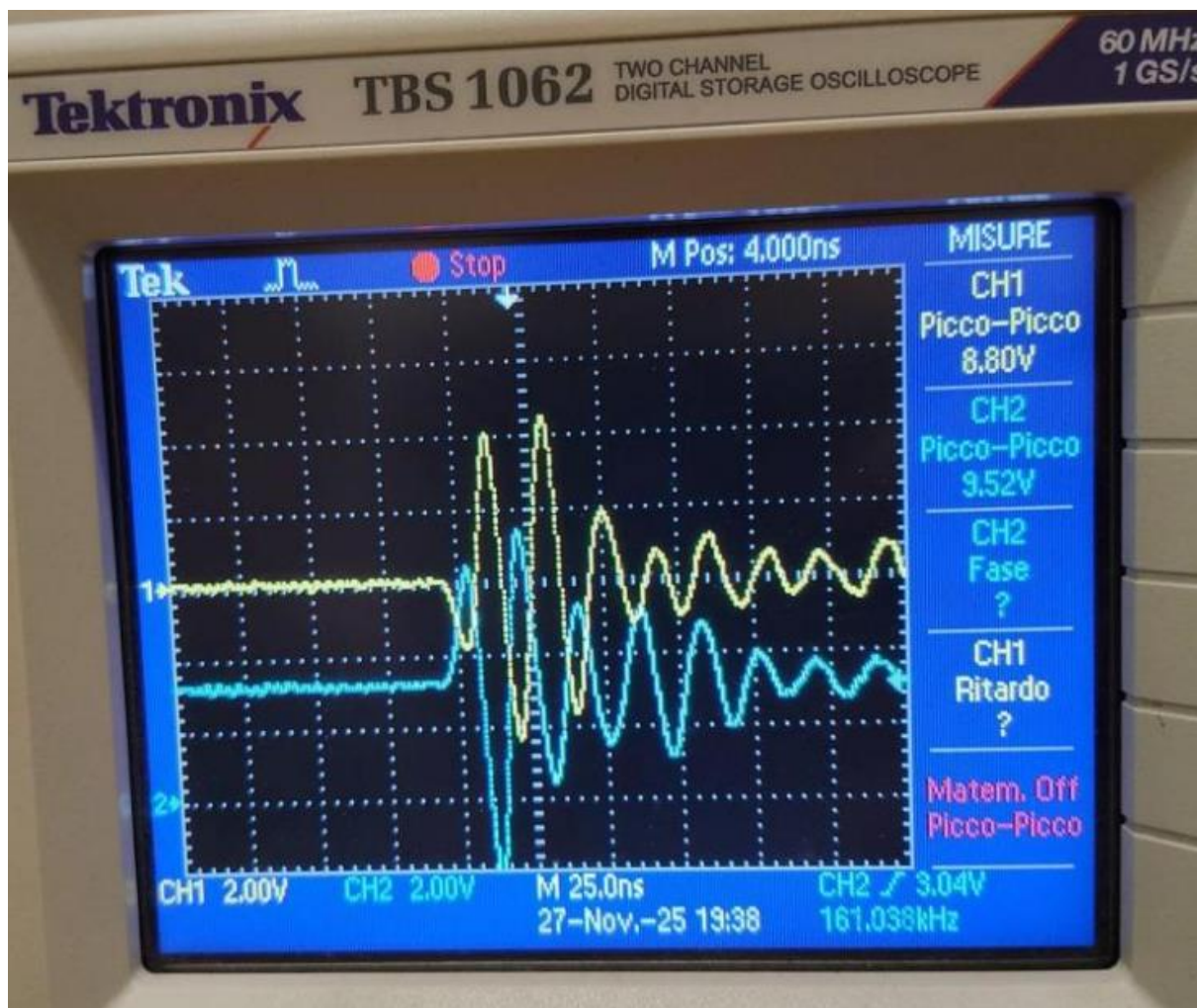
1. Use the oscilloscope to capture screenshots and data files for each measurement.
2. All measurements must reference the ground test point (TP2).
3. Verify the value of the sensing resistor in series with the line ground.
4. Verify that the output jumper is set for 3.3 V operation.
5. Verify the connection of the 10 ohm 5 W resistor as output load.
6. Start with an input voltage of 10 V.

Measurements:

1. Measure the input voltage at the converter input (DC value and AC ripple).
2. Measure the output voltage at the converter output (DC value and AC ripple).
3. Measure the input current using the sensing resistor (DC value and AC ripple).
4. Measure the switching node voltage (diode–mosfet–inductor node) using two time scales to show overshoot and frequency content.
5. Measure the voltage across the inductor terminals (TP4 and TP5) in differential mode.
6. Measure the duty cycle of the converter.
7. Modify the input voltage from 10 V to 15 V in steps of 1 V:
 - 7.1 Record the output voltage and the input current (DC).
 - 7.2 Record the duty cycle.
 - 7.3 Calculate the efficiency (output power/input power).
 - 7.4 Plot output voltage versus input voltage.
 - 7.5 Plot the duty cycle vs input voltage.

The third step was done as follows: after disconnecting the power supply, the input sensing resistance was measured as $R_{in} = 0.5 \Omega$, made of two 1Ω resistors in series. The input current was $I_{in} = 119.6 \text{ A}$. The first tests were done with $V_{in} = 10 \text{ V}$.

Then, the oscilloscope probes were connected between the input and ground (TP2), and between the output and ground (TP2), to record the input and output voltage waveforms.

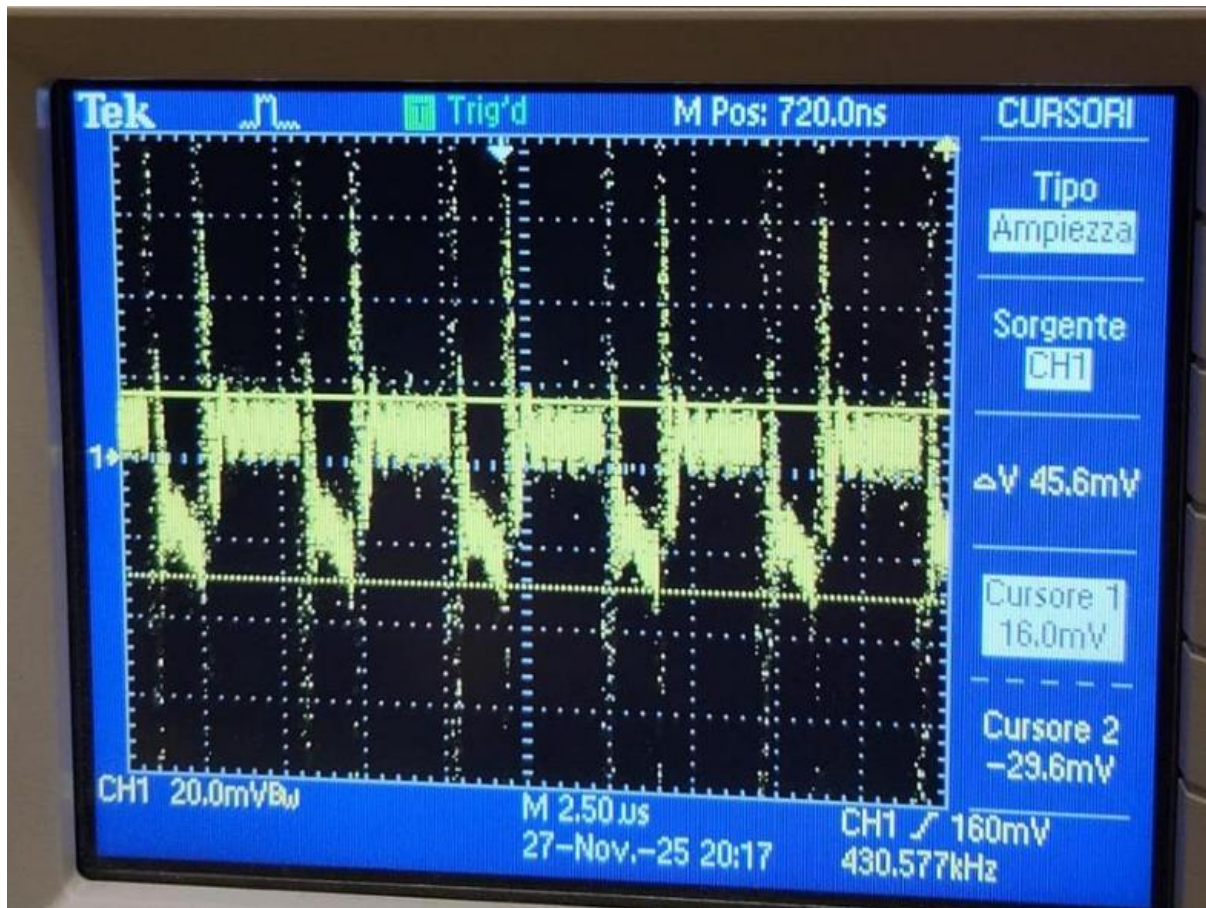


By setting a larger time scale, we measured the DC value and the ripple of the input voltage.

$V_{dc_in} = 10\text{ V}$

$V_{ripple} = 45.6\text{ mV}$

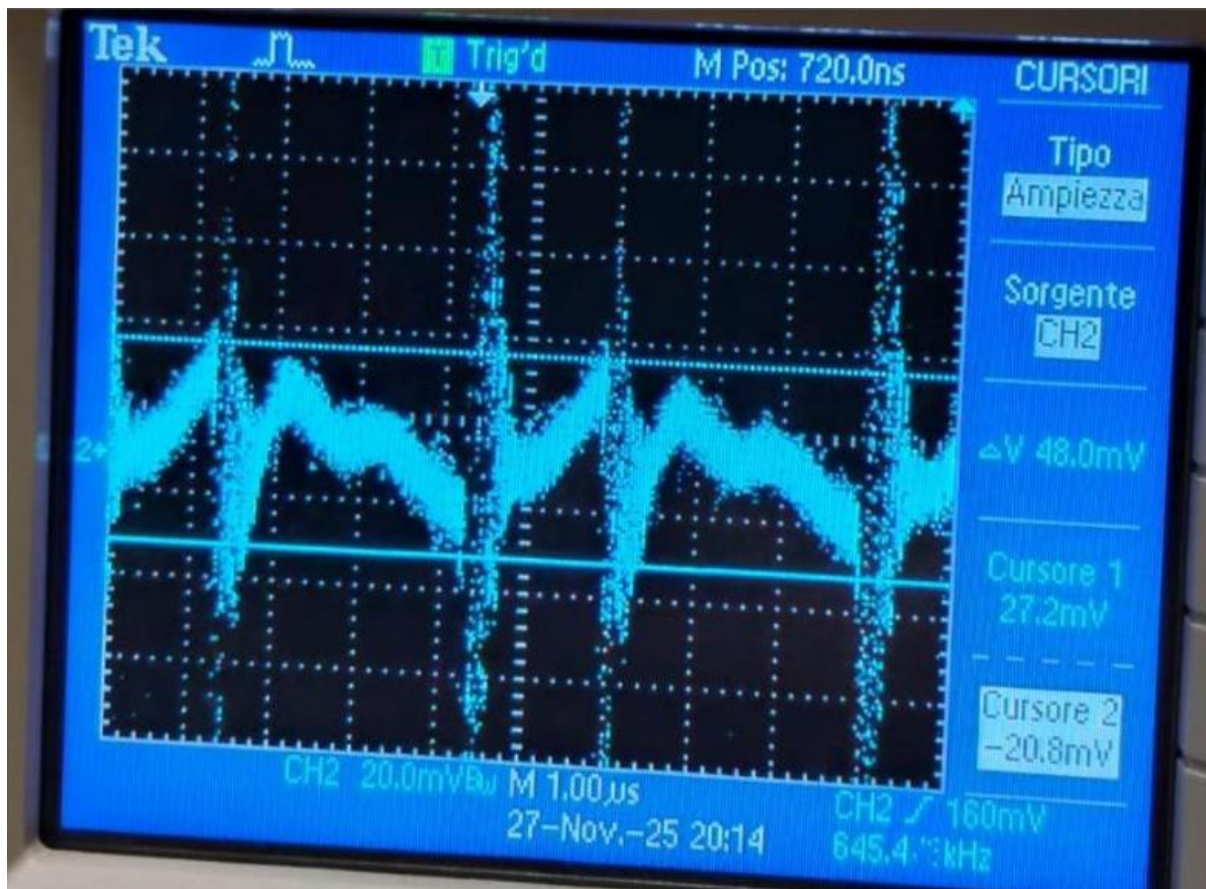
The ripple is caused by the pulsating nature of the input current, and its amplitude is kept low thanks to the input capacitors.



Similarly, for the output voltage:

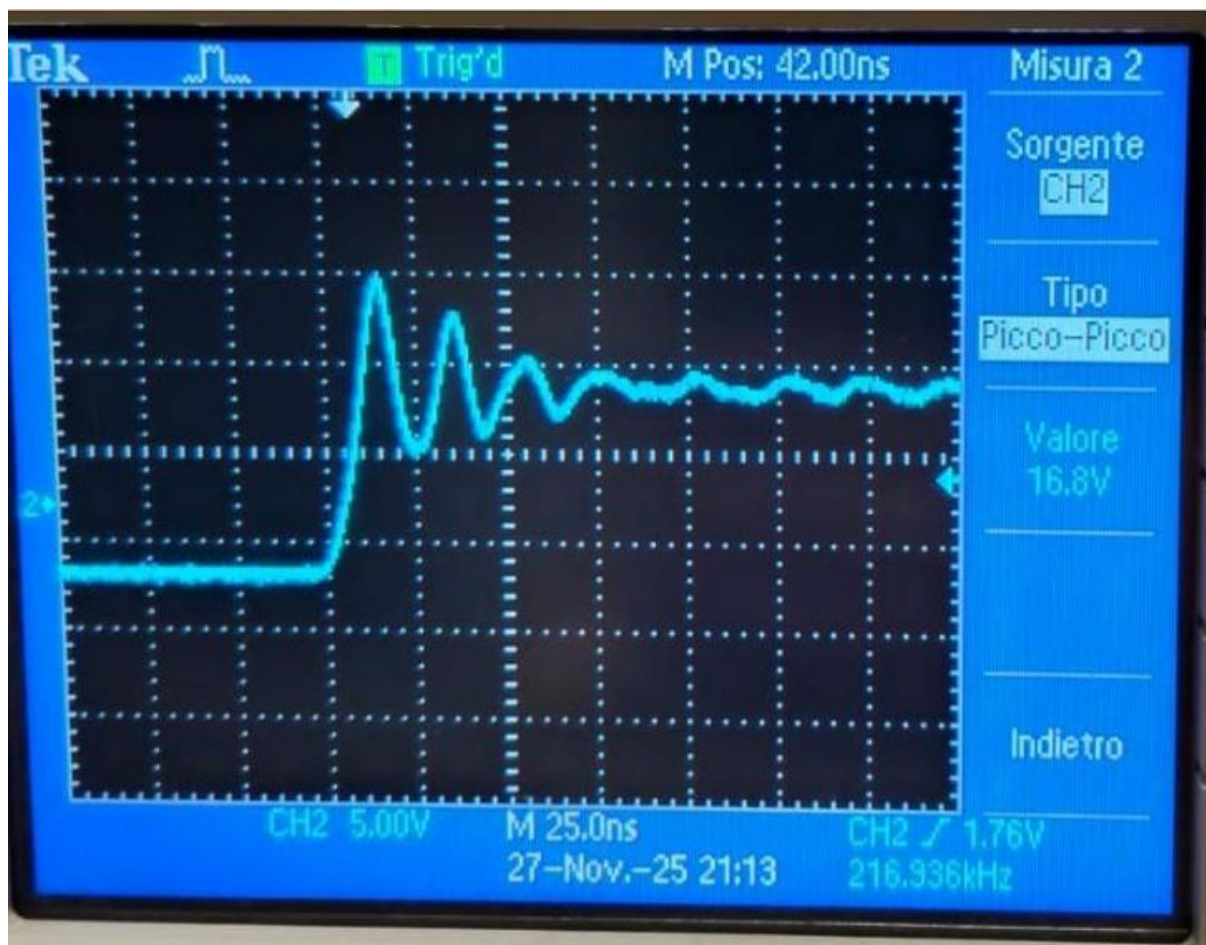
$$V_{dc_o} = 3.17 \text{ V}$$

$V_{ripple_o} = 48 \text{ mV}$, compared to the 31 mV calculated theoretically.



The fourth step was performed by measuring the voltage at the node shared by the diode, MOSFET, and inductor using two different time scales.

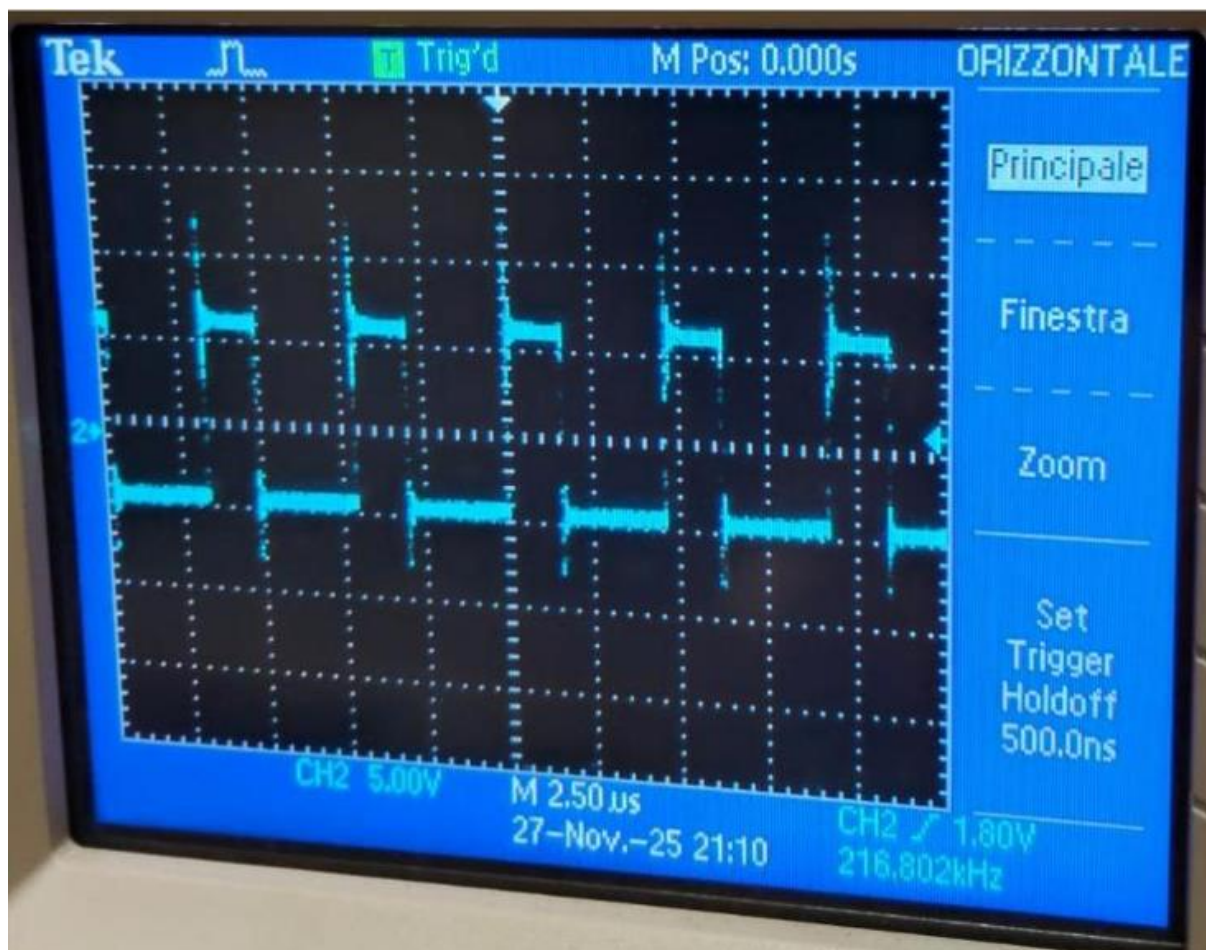
During MOSFET switching, the voltage varies between a maximum value close to V_{in} (minus losses) and 0. When it switches ON, the voltage rises at a rate depending on the MOSFET, which must be slower than the diode recovery, and then stabilizes at the nominal value.



With a larger time scale, it is possible to observe the full MOSFET switching. The switching frequency and duty cycle can be seen.

$D = V_{out} / V_{in} = 0.317$ (point 6). During the ON state, the diode is off and the input voltage is applied to the positive terminal of the inductor. When the MOSFET turns off, the positive terminal is connected to ground through the diode.

Voltage spikes due to parasitic inductances at the measurement node are also visible.



The fifth step requires measuring the differential voltage across the inductor:

$$V_{\text{diff}_1} = 5.13 \text{ V}$$



In the last step, the input voltage was increased from 10 V to 15 V in steps of 1 V. The input current, output voltage, duty cycle variation, and converter efficiency were measured.

V_i	I_i	V_o	D	η
10V	113,6 mA	3,17V	0,317	85%
10,99V	106 mA	3,19V	0,29	87,3 %
12V	99,6 mA	3,2V	0,26	86,66 %
13V	85,6 mA	3,16V	0,24	88 %
14,57V	79,6 mA	3,15V	0,21	85,5 %
15V	73,2 mA	3,11V	0,21	88 %

in conclusion :The output voltage changes only by a few mV as the input voltage varies, so the system is stable and the voltage control loop works properly.

Regarding the input voltage and duty cycle, they are inversely proportional. Since the output voltage is proportional to the input voltage with a gain equal to the duty cycle, the controller reduces the duty cycle at higher V_{in} to keep V_o constant.